AQST| Monday 23/03/2026, US-VNUHCM

AI for quantum
& Quantum for AI
& Quantum AI

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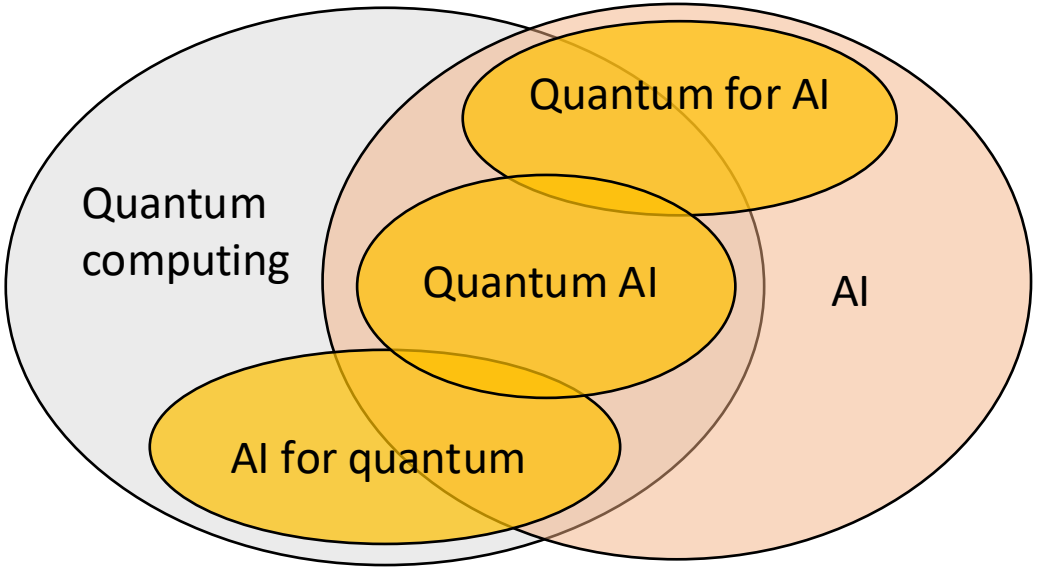
Ms, CS-UIT-VNUHCM, Vietnam: 2023

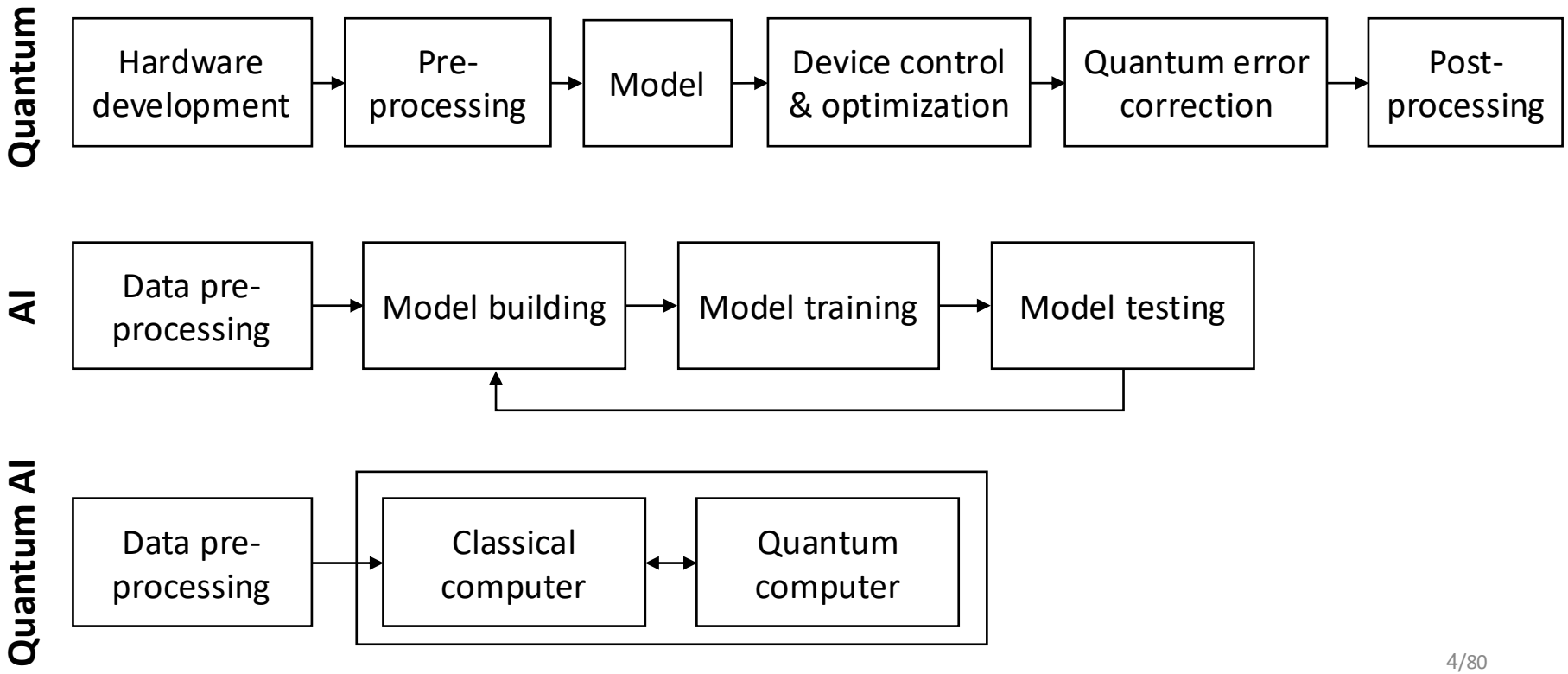
Bs, SE-UIT-VNUHCM, Vietnam: 2021

Outline

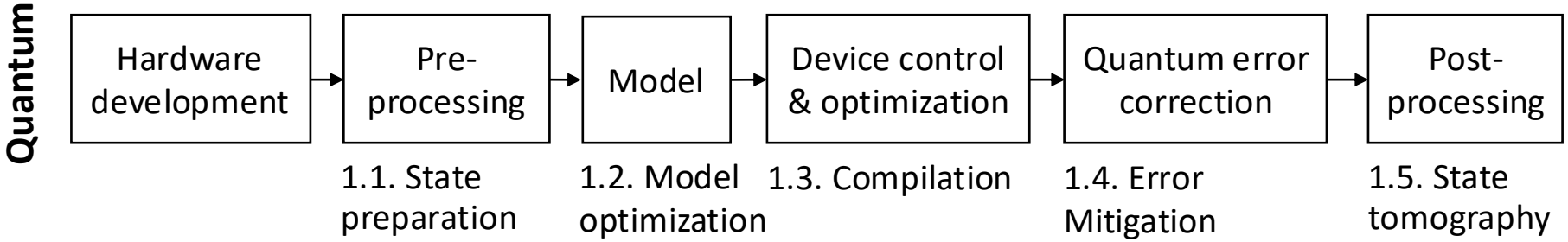
1. AI for quantum
2. Quantum for AI
3. Quantum AI



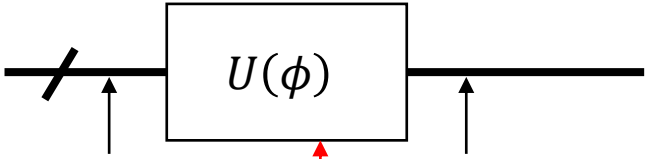
Overview


Quantum
Hardware development
Pre-processing
Model
Device control & optimization
Quantum error correction
Post-processing
AI
Data pre-processing
Model building
Model training
Model testing
Quantum AI
Data pre-processing
Classical computer
Quantum computer

1. AI for quantum



1.1. State preparation



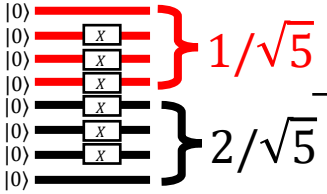
$|\psi_0\rangle = |0\rangle^{\otimes n}$ $|\psi'\rangle = U(\phi)|\psi_0\rangle$ with $|\psi'\rangle$ is the prepared/embedded state

How to prepare $U(\phi)$?

For example: $|\psi'\rangle = \begin{bmatrix} 1/\sqrt{5} & 2/\sqrt{5} \end{bmatrix}$

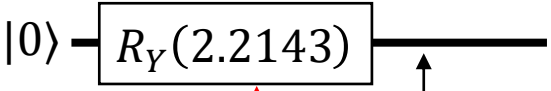
$0.447_{10} \approx 0.0111_2$

$0.8944_{10} \approx 0.1110_2$



$\longrightarrow$ $M2^n$ qubits

Basic encoding



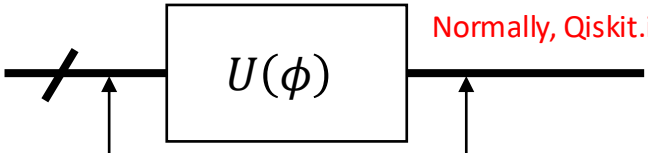
$|\psi'\rangle = \begin{bmatrix} 1/\sqrt{5} \\ 2/\sqrt{5} \end{bmatrix}$ $\longrightarrow$ n qubits

How to find ϕ?

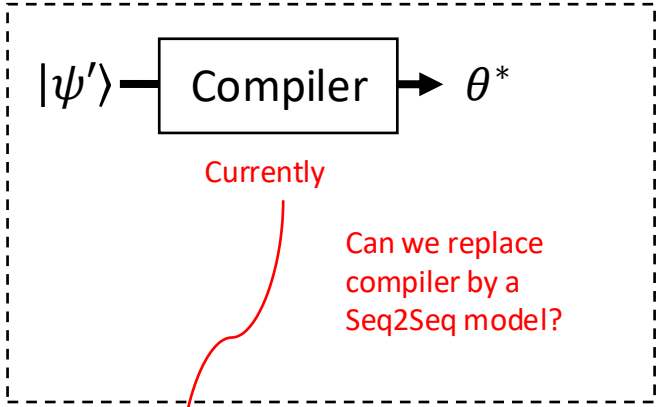
Amplitude encoding

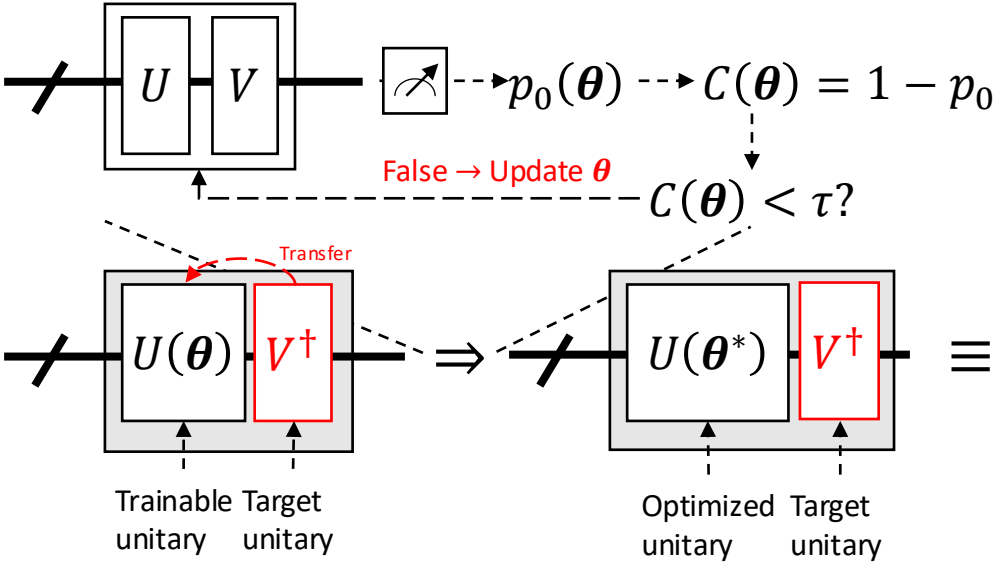
[1.1.1] Schuld, M., & Petruccione, F. (2018). Supervised learning with quantum computers. *Quantum science and technology*, *17*.

1.1. State preparation



Normally, Qiskit.initialize() take exponentially gate & depth for $U(\phi)$! [1.1.2]





[1.1.2] Iten et al., Quantum circuits for isometries (2016). Phys. Rev. A 93, 032318
[1.1.3] Hai, V. T., Viet, N. T., & Ho, L. B. (2024). *SoftwareX*, *26*, 101726. https://doi.org/10.1016/j.softx.2024.101726

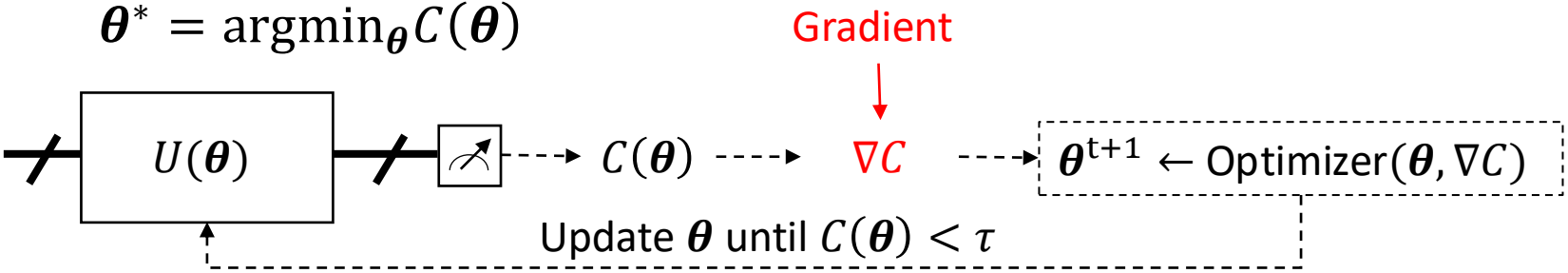
1.2. Model optimization

Considering a PQC $U(\boldsymbol{\theta})$ and cost function:

$$C(\boldsymbol{\theta}) = \langle \mathbf{0} | U^{\dagger}(\boldsymbol{\theta}) \hat{B} U(\boldsymbol{\theta}) | \mathbf{0} \rangle$$

We want to find:

$$\boldsymbol{\theta}^{*} = \mathrm{argmin}_{\boldsymbol{\theta}} C(\boldsymbol{\theta})$$



1.2.1. Gradient

Computing PQC's gradient is the key in quantum optimization:

$$\nabla C = [\partial_{\theta_0} C \quad \dots \quad \partial_{\theta_{m-1}} C]^{\intercal}$$

The partial derivative can be computed through parameter-shift rule **[1.17]**:

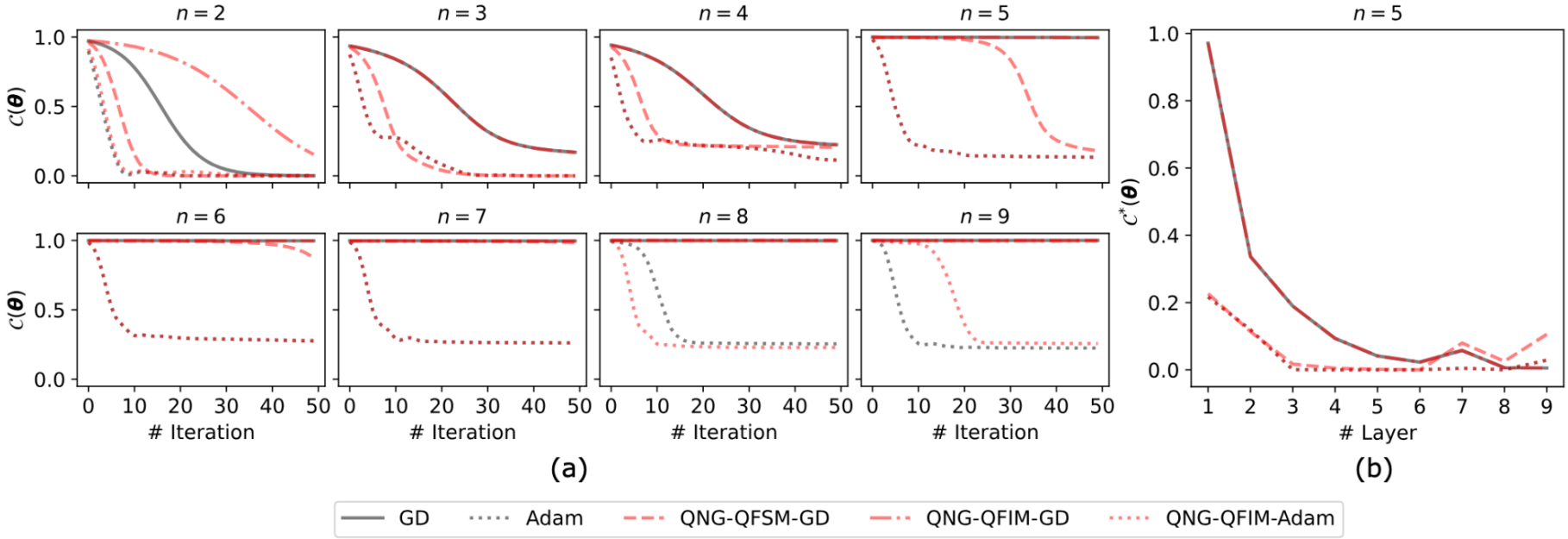
$$\frac{\partial C}{\partial \theta_j} = \sum_{k=1}^{R} d_k \left(C\left(\theta_j^{+\epsilon_k}\right) - C\left(\theta_j^{-\epsilon_k}\right)\right)$$

and $C\left(\theta_j^{+\epsilon_k}\right) \equiv C(\boldsymbol{\theta} + \mathbf{e_j}\epsilon_k)$, $\mathbf{e_j}$ is the jth unit vector, ϵ_k is the shift-value

[1.2.1] D. Wierichs, J. Izaac, C. Wang, and C. Y.-Y. Lin, "General parameter-shift rules for quantum gradients," Quantum, vol. 6, p. 677, Mar. 2022

1.2.2. Optimizer

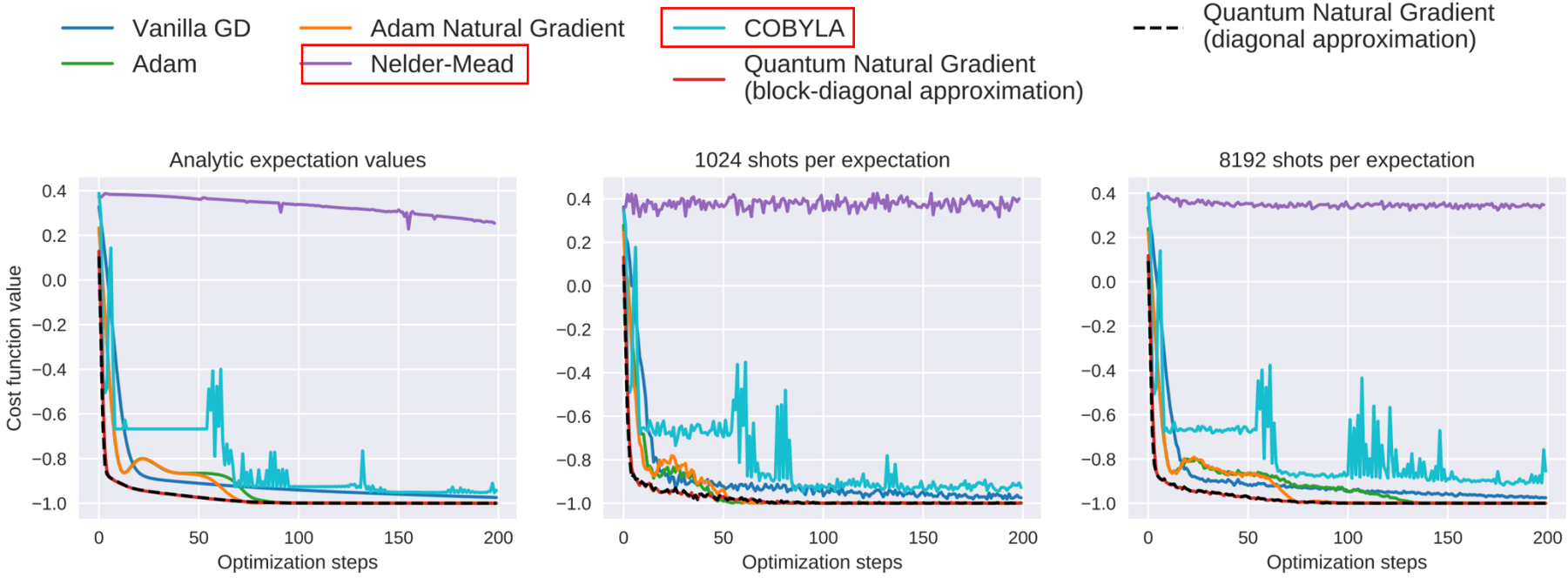
Quantum optimization is still heavily depend on classical optimizers such as Adam



[1.2.2] Hai, V. T., Luan, P. H., & Nakashima, Y. (2024, November). Benchmarking Classical and Quantum Optimizers for Quantum Simulator. In *2024 Twelfth International Symposium on Computing and Networking (CANDAR)* (pp. 238-244). IEEE.

1.2.2. Optimizer

Note that we can also use gradient-free optimizers: Nelder-Mead, COBYLA [**1.2.3**]



[1.2.3] Stokes, J., Izaac, J., Killoran, N., & Carleo, G. (2020). Quantum natural gradient. *Quantum*, *4*, 269.

1.3. Compilation

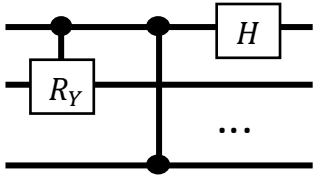
There is a large gap between algorithms and set of instructions we can perform on real qc

Problem 1 (a): The native gate set is diff

No CX, R_X or CR_Y — $\{CZ, X, R_Z, SX, ID\}$ — IBM series

No CX, R_Y or CR_Y — $\{R_X, R_Z, ISWAP\}$ — Rigetti Ankaa-3

If we have this example circuit:



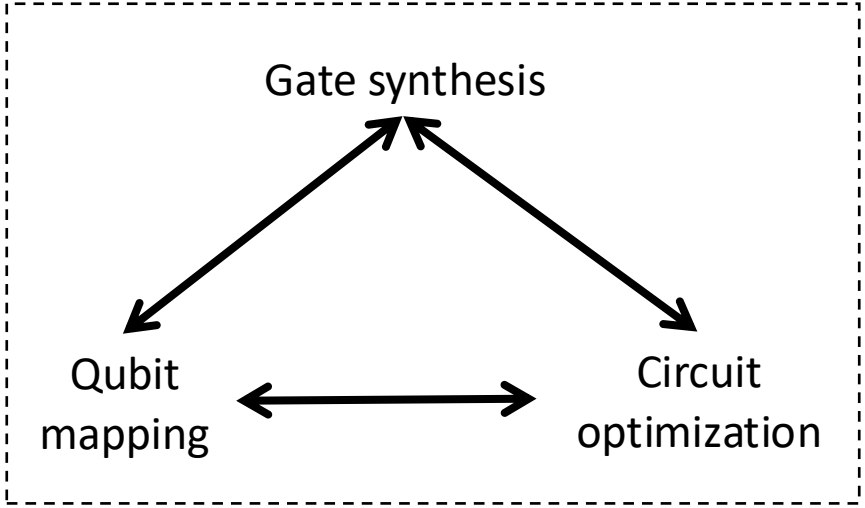
$\equiv$ $U_{\{u\}}$

$\Downarrow$ Compile

$U'_{\{u'\}}$

$$u = \{R_{X,Y,Z}, CX\}$$

$$u' = \{R_{X,Z}, ISWAP\}$$



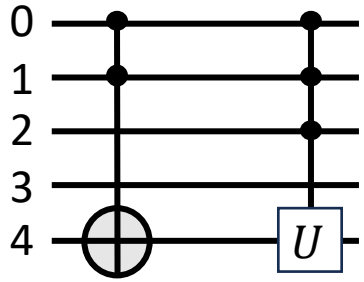
Quantum compilation stage

[1.3.1] Bäumer, E., Tripathi, V., Wang, D. S., Rall, P., Chen, E. H., Majumder, S., ... & Minev, Z. K. (2024). Efficient long-range entanglement using dynamic circuits. *PRX Quantum*, *5*(3), 030339.

1.3. Compilation

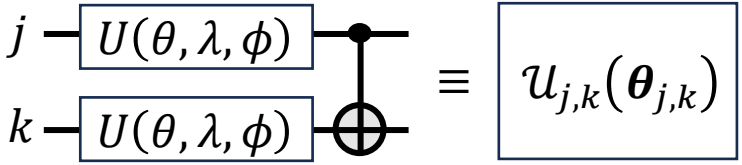
Gap between algorithms and set of instructions we can perform on real qc

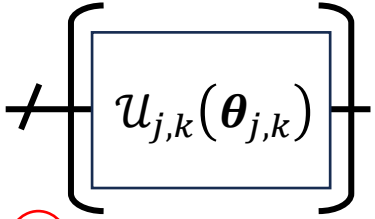
Problem 1 (b): Some gates are not exist



Toffoli gate Multi-control gate

- Native gate set $\{\boldsymbol{u}'\}$
- Qubit layout





Find 'active' blocks and optimal $\boldsymbol{\theta}^*$

$U'_{\{\boldsymbol{u}'\}}$

n^2 blocks for all possible $\{j,k\}$

Number of qubits	2	3	4	5	n_{CX}
Iterative unentangling [21]	8	62	344	1642	$2\cdot 4^n-(2n+3)\cdot 2^n+2n$
Givens rotations [20, 27]	4	64	536	4156	$\approx 8\cdot 4\cdot 4^n$
Recursive CSD [22]	14	92	504	2544	$\frac{1}{2}\cdot n\cdot 4^n-\frac{1}{2}\cdot 2^n$
Recursive CSD (optimized) [23]	4	26	118	494	$\frac{1}{2}\cdot 4^n-\frac{1}{2}\cdot 2^n-2$
QSD [21, 27]	6	36	168	720	$\frac{3}{4}\cdot 4^n-\frac{3}{2}\cdot 2^n$
QSD (optimized)[21]	3	20	100	444	$(23/48)\cdot(4^n)-\frac{3}{2}\cdot 2^n+\frac{4}{3}$
Sequential optimization[a]	**3**	**15**	**63**	**267**	Not available
Theoretical Lower Bounds[18]	3	14	61	252	$\frac{1}{4}(4^n-3n-1)$

Note that there is a theoretical limit (lower bound)

SQUANDER package **[1.13]**

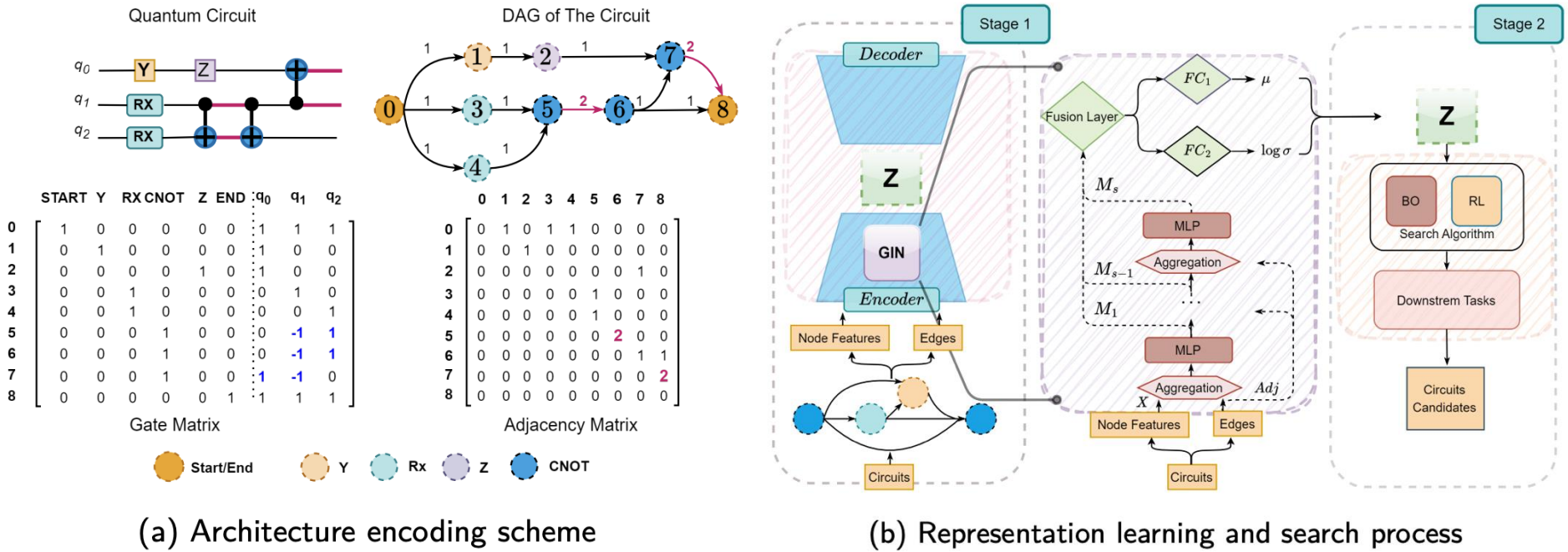
Reduce this complexity by AI models?

[**1.3.2**] Rakyta, P., & Zimborás, Z. (2022). Approaching the theoretical limit in quantum gate decomposition. *Quantum*, *6*, 710.

1.3. Compilation

Can we speed-up compilation process by AI models?

- Quantum architecture search (finding the best ansatz) **[1.3.3]**



(a) Architecture encoding scheme

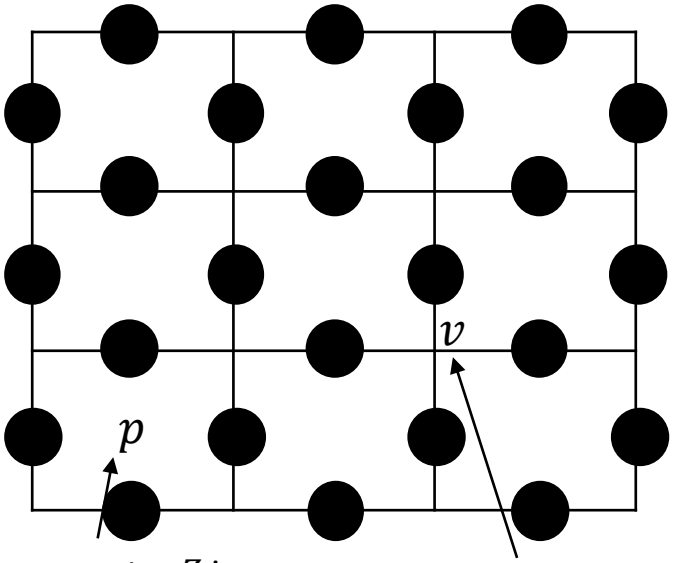
(b) Representation learning and search process

[1.3.3] Quantum Architecture Search with Unsupervised Representation Learning, Yize Sun, Zixin Wu, Volker Tresp, and Yunpu Ma, Quantum 10, 1994 (2026).

1.4. Quantum error correction

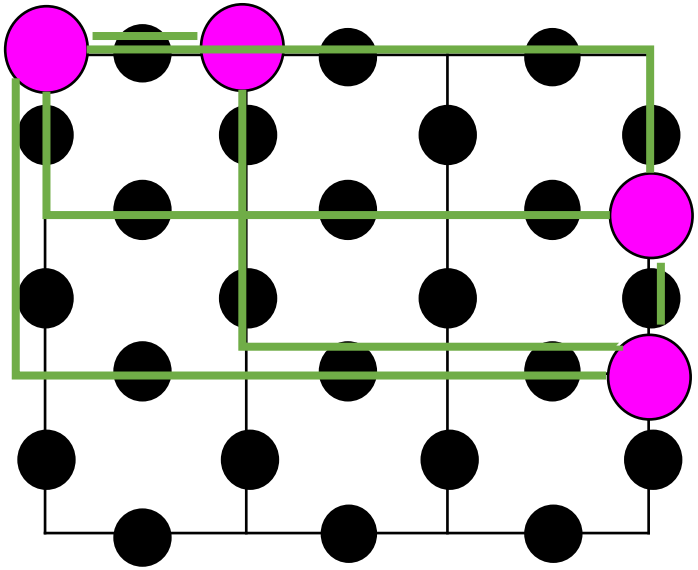
MWPM decoder (Minimum Weight Perfect Matching) used in Surface code

syndrome → error qubits



Stabilizer operator Z is surrounded by qubitS

Stabilizer operator X

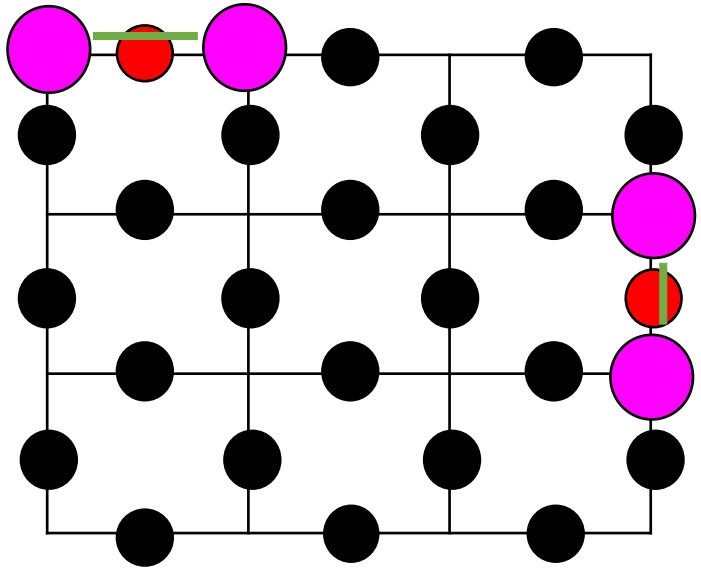


Detected syndrome!

1.4. Quantum error correction

MWPM decoder (Minimum Weight Perfect Matching)

Find shortest pairs through Blossom algorithm

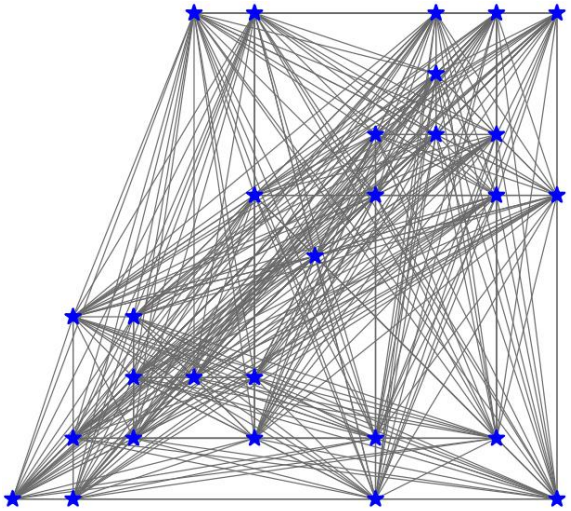


Matched

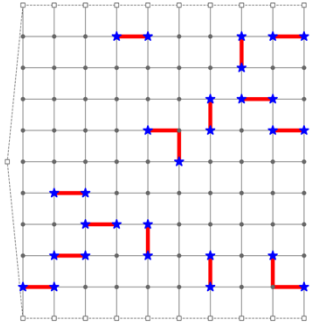
1.4. Quantum error correction

Complexity: $O(N^3 logN)$ with N is the number of syndrome.

How to reduct it?



$\Rightarrow$



CNN: look graph, then guess the ver/horiz line

Graph NN: perfect candidate?

Transformer: ???

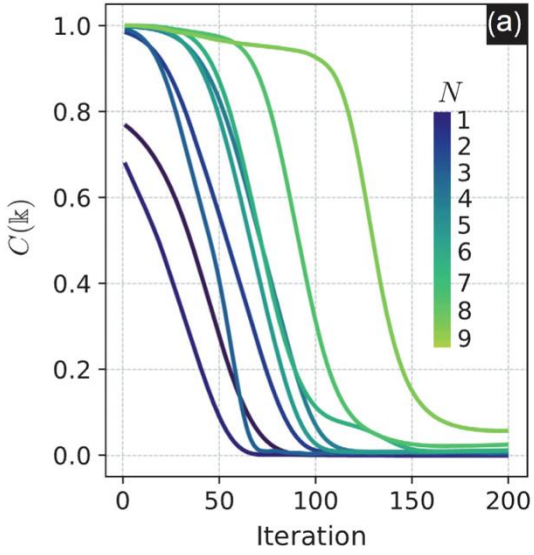
[1.4.1] Higgott, O. (2022). Pymatching: A python package for decoding quantum codes with minimum-weight perfect matching. *ACM Transactions on Quantum Computing*, *3*(3), 1-16.

1.5. Tomography

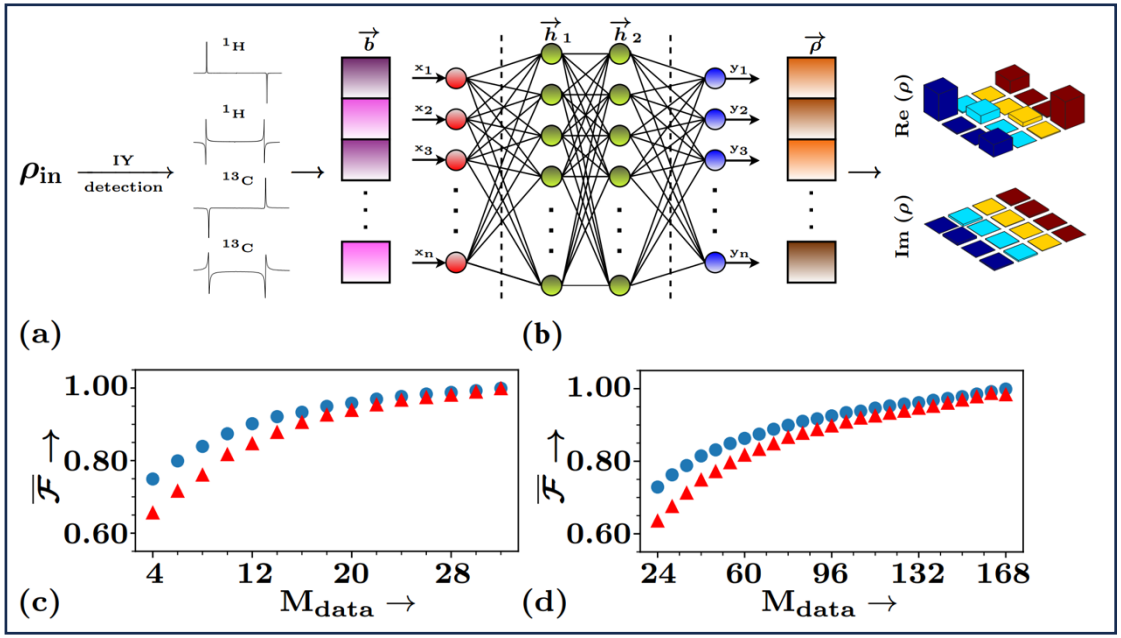
We have an unknown quantum channel $\mathcal{E}$ affect on ρ_{in} st $\rho_{out} = \mathcal{E}(\rho_{in})$

$\mathcal{E}(\rho) = \sum_{i,j} \mathcal{X}_{i,j} K_i \rho K_j^{\dagger}$ where $\{K\}$ is the basis operators, $\mathcal{X}$ is the learnable operator

$\Rightarrow$ We want to find $\mathcal{X}$ through 2^n measurements. Instead of that,



(1) We achieve it by an optimization process.



(2) By an neural net

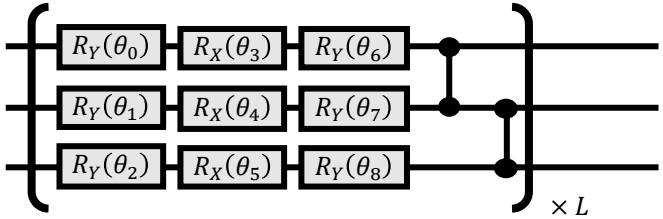
[1.5.1] Linh, H. L. D., Hai, V. T., & Ho, L. B. (2026). Advancing Quantum Process Tomography Through Quantum Compilation. *Advanced Quantum Technologies*, *9*(2), e00494.

[1.5.2] Gaikwad, A., Bihani, O., Arvind, & Dorai, K. (2024). Neural-network-assisted quantum state and process tomography using limited data sets. Physical Review A, 109(1), 012402.

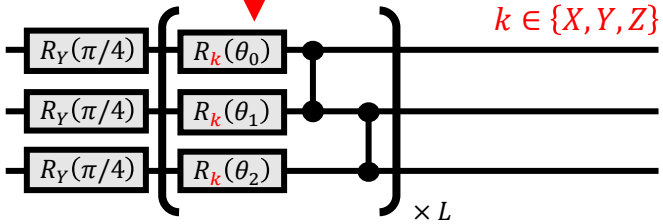
1.6. Additional informations

Barren platueas (BP) → A method for mitigate BP

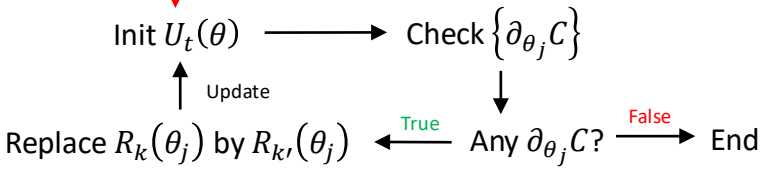
Different ansatz affect strongly to the performance

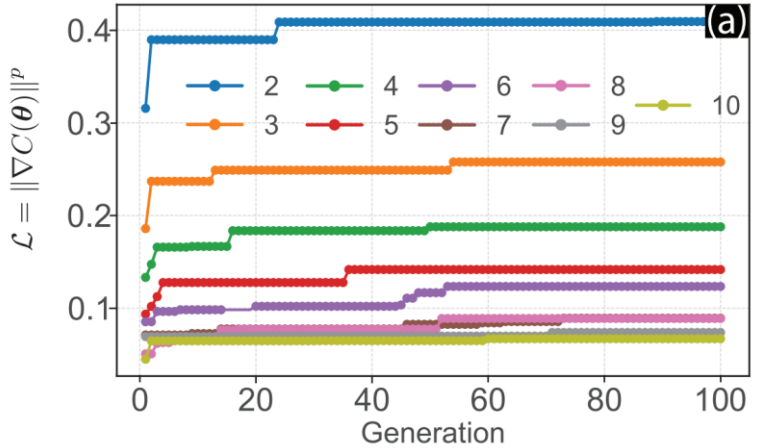


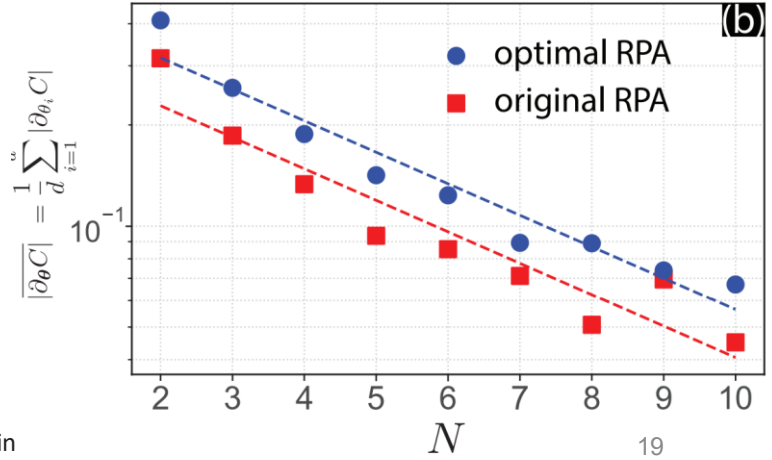
Hardware-effecitent ansatz (HEA)



Random Pauli ansatz (RPA)







[1.6.1] Ho, L. B., Urbaneja, J., & Ashhab, S. (2025). Statistical analysis of barren plateaus in variational quantum algorithms. *arXiv preprint arXiv:2508.08915*.

1.6. Additional informations

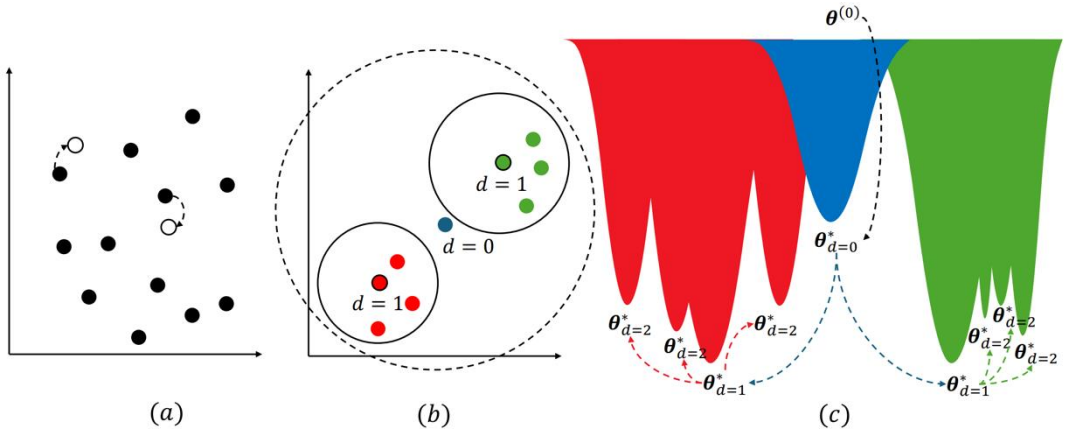
Can we apply classical techniques into quantum optimization?

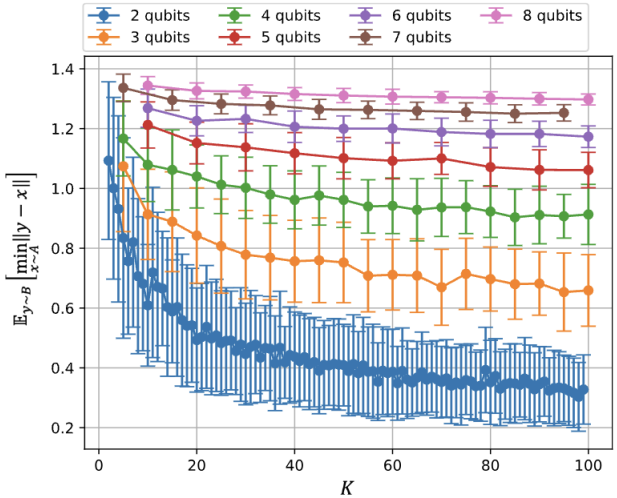
Consider a set of K optimization targets denoted as $T_1, T_2, \dots, T_K$, all defined over the same search space S. Each target T_k is associated with its own cost function:

$$C\left(\theta_{(k)}\right): R^m \rightarrow [0,1]$$

The goal of multi-target optimization is to find a set of solutions:

$$\{\theta^*_{(1)}, \theta^*_{(2)}, \dots \theta^*_{(K)}\}$$





[1.6.2] Hai, V. T., Doanh, B. C., Duong, L. V. T., Luan, P. H., & Nakashima, Y. (2025). Transfer-Based Strategies for Multi-Target Quantum Optimization. *CANDAR 2025.*

1.6. Additional informations

Can we apply classical techniques into quantum optimization?

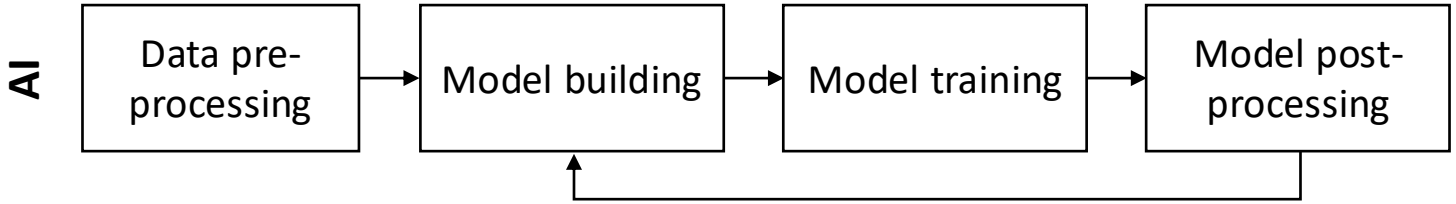
Warm-starting method: simply set $\theta^*_{x_{(k)}}$ as initial parameter $\theta^{(0)}_{y_{(k')}} = \theta^*_{x_{(k)}}$.

If $y \approx x + \epsilon$, x and y are targets, considering the first-order Taylor expansion of cost function C with respect to $\boldsymbol{\theta}$ at $\boldsymbol{\theta}^*_{\boldsymbol{x}}$ and fixed y, we have:

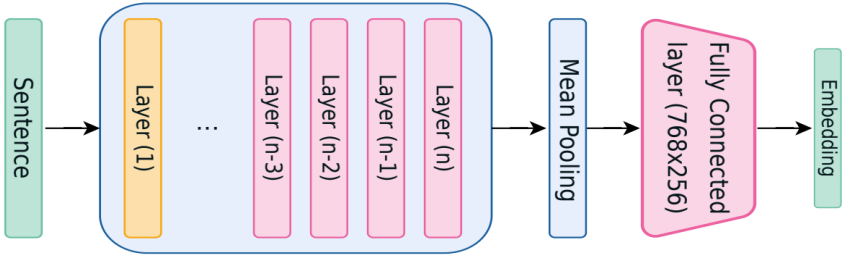
$$\tilde{\boldsymbol{\theta}}^*_y \approx \boldsymbol{\theta}^*_x - \frac{\nabla_\theta C_y(\theta^*_x)}{\left\|\nabla_\theta C_y(\theta^*_x)\right\|^2} C_y(\theta^*_x)$$

Parameter estimator

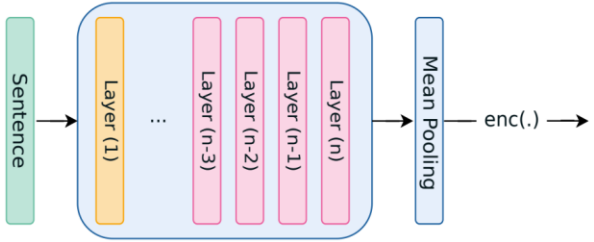
2. Quantum for AI (Quantum-inspired model)

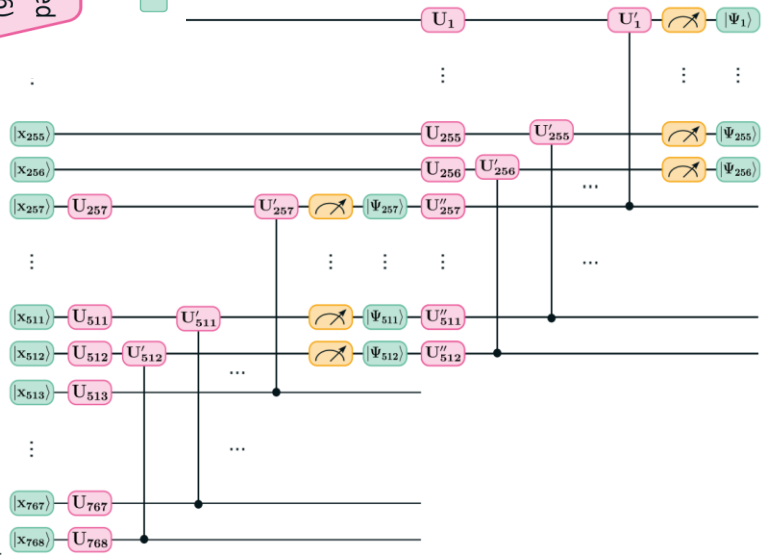


2.1. Data pre-processing: compression task



[x] Classical embedding compression for the base BERT model. Only the red components are tunable.

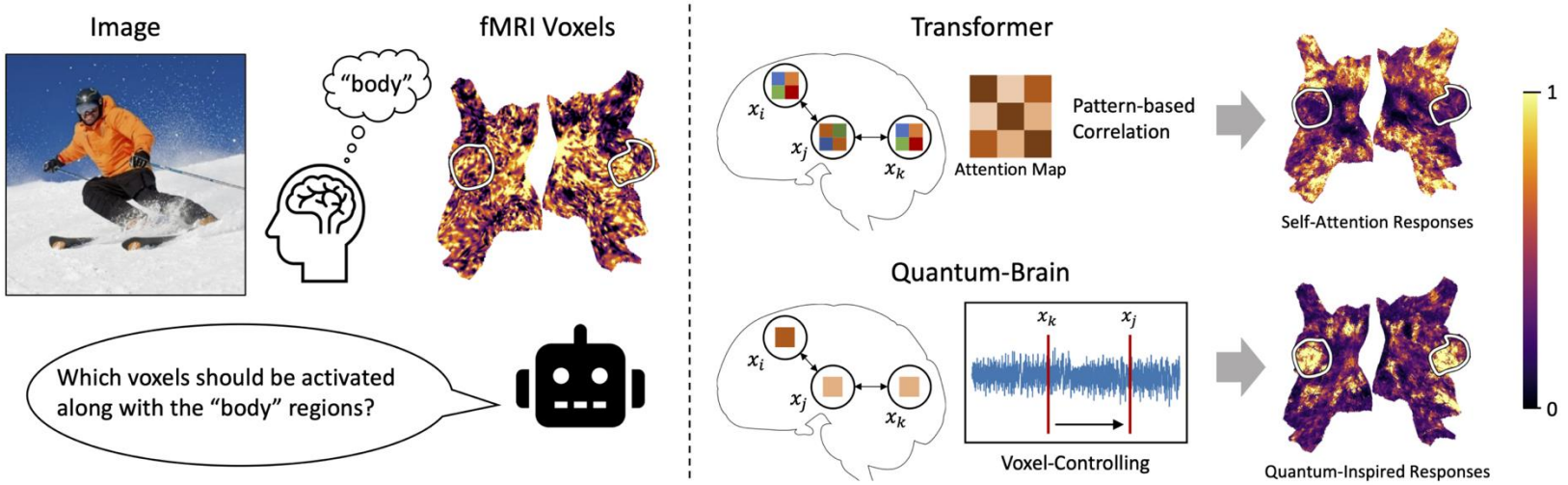




Quantum-inspired embedding compression of base BERT. x32 fewer parameters???

[2.1.1] Kankeu, I., Fritsch, S. G., Schönhoff, G., Mounzer, E., Lukowicz, P., & Kiefer-Emmanouilidis, M. (2025). Quantum-inspired embeddings projection and similarity metrics for representation learning. *arXiv preprint arXiv:2501.04591*.

2.2. Model building: quantum-inspired model



Quantum-Brain for Vision-Brain Understanding. Motivated by the connectivities in brain signals and entanglement in quantum theory, we propose Quantum-Brain, a quantum-inspired neural network to explore the connectivities in brain signals for vision-brain understanding. Apart from self-attention, we evaluate the entanglement between brain voxels to enhance the exploration of connectivity. **[2.2.1]**

[2.2.1] Nguyen, H. Q., Nguyen, X. B., Churchill, H., Choudhary, A. K., Sinha, P., Khan, S. U., & Luu, K. (2024). Quantum-brain: Quantum-inspired neural network approach to vision-brain understanding. arXiv preprint arXiv:2411.13378.

2.2. Model building: quantum-inspired model

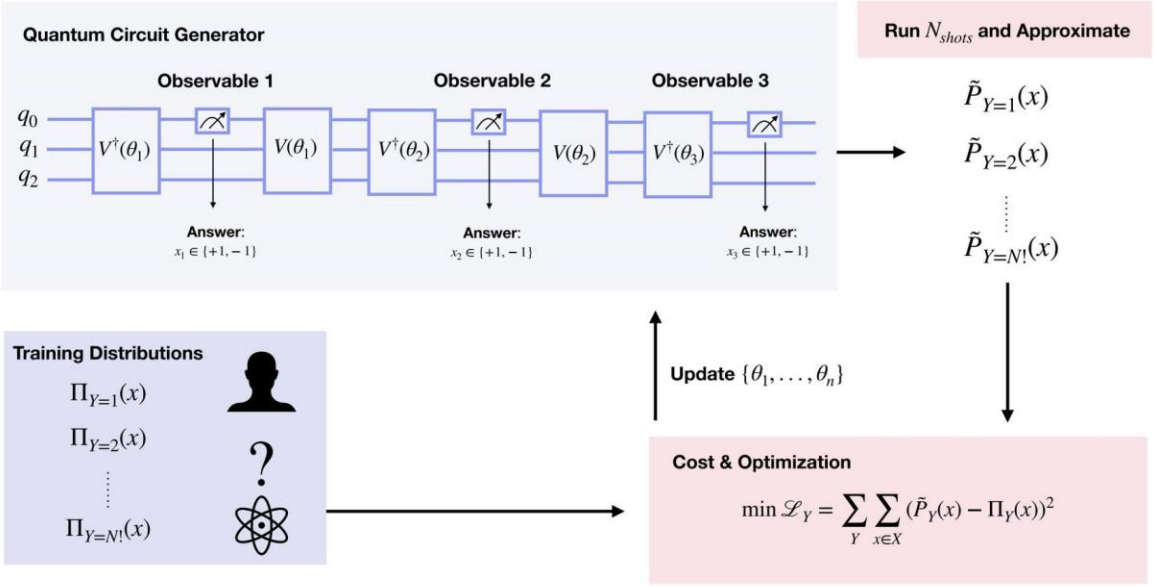
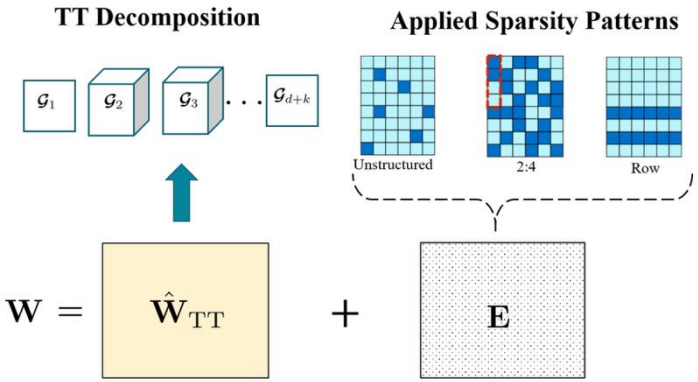


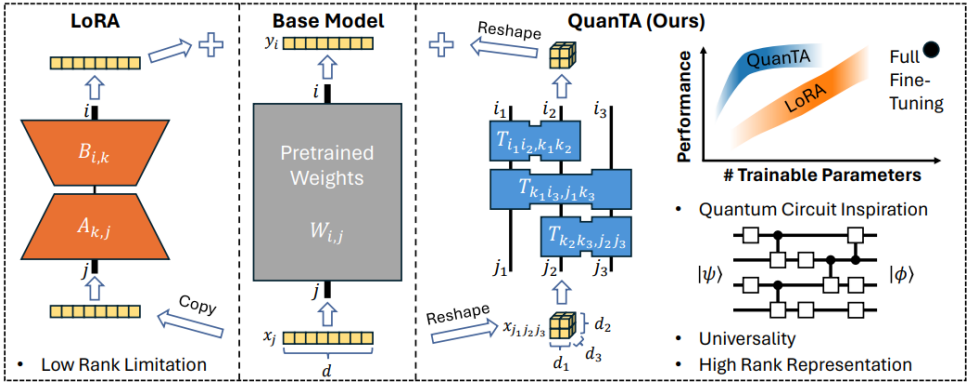
Fig. 1 An example of the multi-task quantum generative architecture for training on binary distributions with three observables. We provide a visual overview of the quantum circuit generator, containing subsequent measurement observables of the form $V(\theta_n) D_n V(\theta_n)^\dagger$. Measurements taken in the Pauli-Z basis allow one to obtain a binary eigenvalue (answer) for each trainable observable. The generator is run for a number of N_{shots} to approximate the desired number of model output distributions $\tilde{P}_Y(x)$ over binary answer variables, with respect to the order sequence of the observables. Swapping the order sequence simply corresponds to physically swapping the parameters θ_n in the generator, which are indexed by Y. The model is trained to minimize the multi-task loss function $\mathcal{L}_Y$ that computes the LSM loss on the training distribution $\Pi_Y(x)$ and the empirical model distribution $\tilde{P}_Y(x)$ for the desired number of tasks to use for training

[2.2.2] Gili, K., Alonso, G., & Schuld, M. (2024). An inductive bias from quantum mechanics: learning order effects with non-commuting measurements. Quantum Machine Intelligence, 6(2), 67

2.3. Post-processing: compressing by Tensor-network



(A) The sparse + low-rank tensor train representation for a weight matrix (or embedding table). **[2.3.1]**



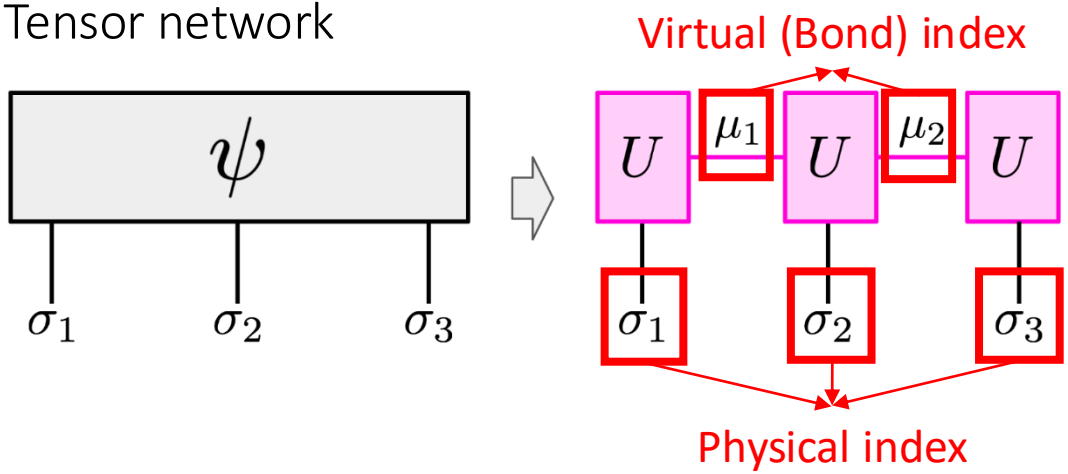
(B) Conceptual comparison of QuanTA and LoRA methods. LoRA parameterizes the weight matrix update as a outer product of two low-rank matrices, limiting its capacity. QuanTA, inspired by quantum circuits, uses tensors that operate on specific axes of the (reshaped) input, enabling high-rank parameterization.**[2.3.2]**

[2.3.1] Solgi, R., Zhen, K., Swaminathan, R. V., Susanj, N., Mouchtaris, T., Kunzmann, J., & Zhang, Z. (2025). *Sparse augmented tensor networks for post-training compression of large language models*. EMNLP2025

[2.3.2] Chen, Z., Dangovski, R., Loh, C., Dugan, O., Luo, D., & Soljačić, M. (2024). QuanTA: Efficient High-Rank Fine-Tuning of LLMs with Quantum-Informed Tensor Adaptation. In A. Globerson, L. Mackey, D. Belgrave, A. Fan, U. Paquet, J. Tomczak, & C. Zhang (Eds), Advances in Neural Information Processing Systems (Vol. 37, pp. 92210–92245). doi:10.52202/079017-2928

2.3. Post-processing: tensor network

Tensor network



We present 2^n statevector as $\underbrace{(2,2,\dots,2)}_{n}$ tensor

For qubit 0, we reshape $|\psi\rangle$ as row = (i_1), col = $(i_2,\dots,i_n)$, $\psi_{i_1,(i_2,\dots,i_n)} \xrightarrow{SVD} \boxed{A_1^{i_1}}\boxed{U}\boxed{SV^\dagger}$

For qubit 1, we reshape $SV^\dagger$ as row = $(bond \times i_2)$, col = $(i_3,\dots,i_n)$ $\xrightarrow{SVD} A_2^{i_2} SV^\dagger$

…

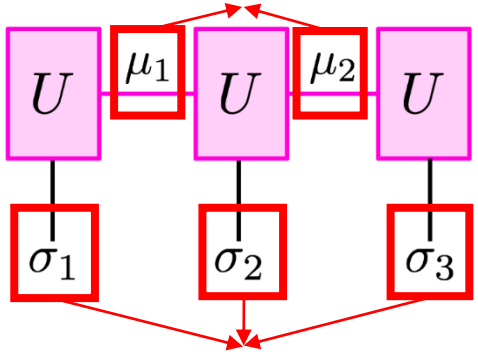
Finally, we collect $\left\{A_1^{i_1}, A_2^{i_2}, \dots, A_n^{i_n}\right\}$

2.3. Post-processing: tensor network

Tensor network

$$|W_4\rangle = 1/\sqrt{4}(|0001\rangle + |0010\rangle + |0100\rangle + |1000\rangle)$$

Virtual (Bond) index



Physical index

In qubits system: $\sigma_j = \{0,1\}$

$$|W_4\rangle \xrightarrow{(lvbond\times|\sigma_1|)\times\frac{2^n}{lvbond\times|\sigma_1|}} \psi_{i_1,(i_2 i_3 i_4)} \xrightarrow{SVD} U\Sigma V^\dagger$$

U Σ $V^\dagger$

2×2 2×2 2×8

$$\Sigma = \begin{bmatrix} \sqrt{3}/2 & 0 \\ 0 & 1/2 \end{bmatrix}$$

(l)vbond = 1

(r)vbond = {1,2}

$$U(\mu_0,\sigma,\mu_1) \rightarrow \begin{cases} A_{0,\mu_0,\mu_1} = \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} \end{bmatrix} \\ A_{1,\mu_0,\mu_1} = \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{-2}{\sqrt{6}} \end{bmatrix} \end{cases}$$

For next iteration $\boxed{B = \Sigma V^\dagger}$

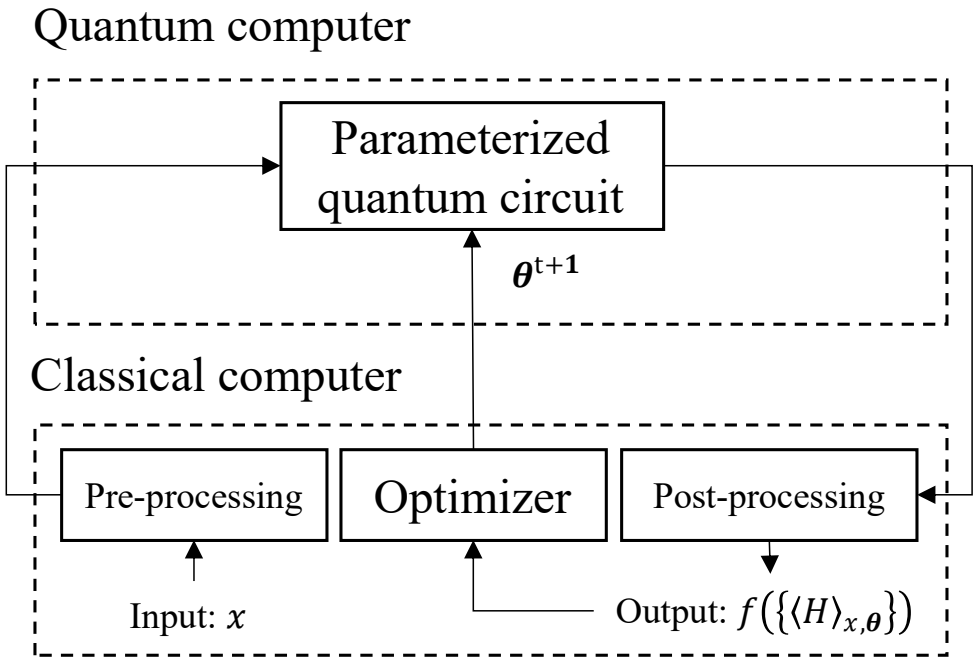
Collect A_{σ_0,μ_0,μ_1}

[2.3.3] Siddhartha Patra, Saeed S. Jahromi, Sukhbinder Singh, Roman Orus "Efficient tensor network simulation of IBM's largest quantum processors" arXiv:2309.15642, 2023.

[2.3.4] H. C. Jiang, Z. Y. Weng, T. Xiang "Accurate determination of tensor network state of quantum lattice models in two dimensions" arXiv:0806.3719, 2008.

3. Quantum AI

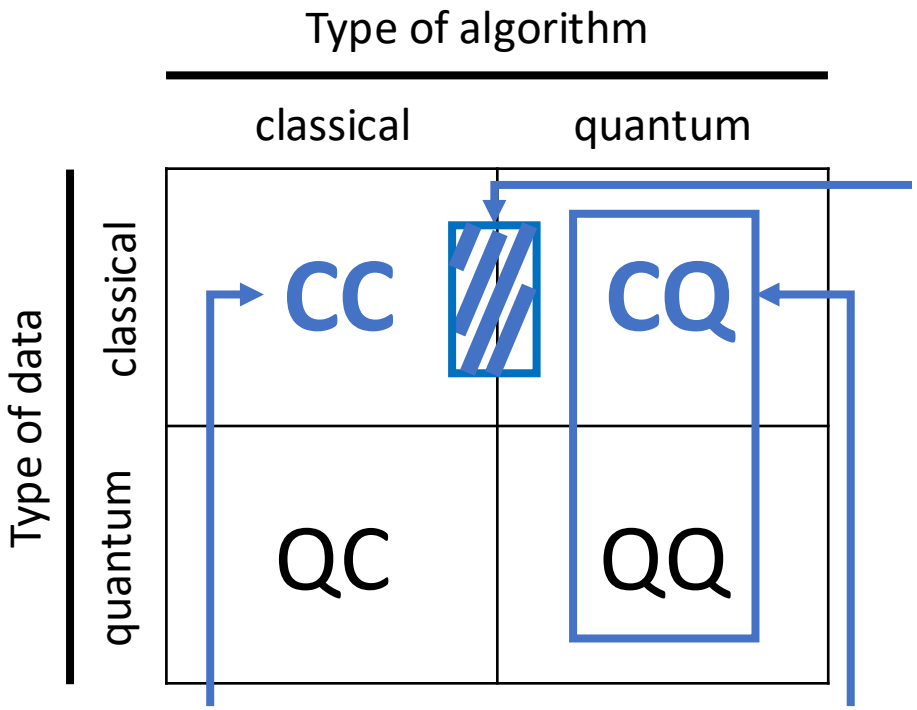
Hybrid quantum - classical model



Hybrid model [3.0.1]

[3.0.1] M. Schuld and F. Petruccione, Machine learning with quantum computers (Springer, 2021)

3. Overview



Hybrid quantum – classical learning: algorithm running on both quantum and classical computers

QML model should resolve below classical machine learning problems:

(1) Require huge dataset
(2) High computation resources
(3) Lack of generality

Classical model: KNN, SVM, Deep learning, …

Quantum Machine learning (QML): algorithm running on only quantum computer or both quantum / classical computers

3. Overview

What QML models can do?
Based on quantum advantage [3.0.3]

(1) Require huge dataset

⇒ Require less items

(2) High computation resources

⇒ Lower time complexity

(3) Lack of generality

⇒ Example: Require less #parameter than classical ML model or achieve higher accuracy on the same dataset.

Be careful with hype QML papers!

[3.0.2] Simões, R. D. M., Huber, P., Meier, N., Smailov, N., Füchslin, R. M., & Stockinger, K. (2023). Experimental evaluation of quantum machine learning algorithms. *IEEE Access*, *11*, 6197-6208.
[3.0.3] Schuld, M., & Killoran, N. (2022). Is quantum advantage the right goal for quantum machine learning?. *Prx Quantum*, *3*(3), 030101.

3. Overview

Type of QML model

a) Speed up version of classical ML algorithms:

- QKNN
- QKmean
- QSVM

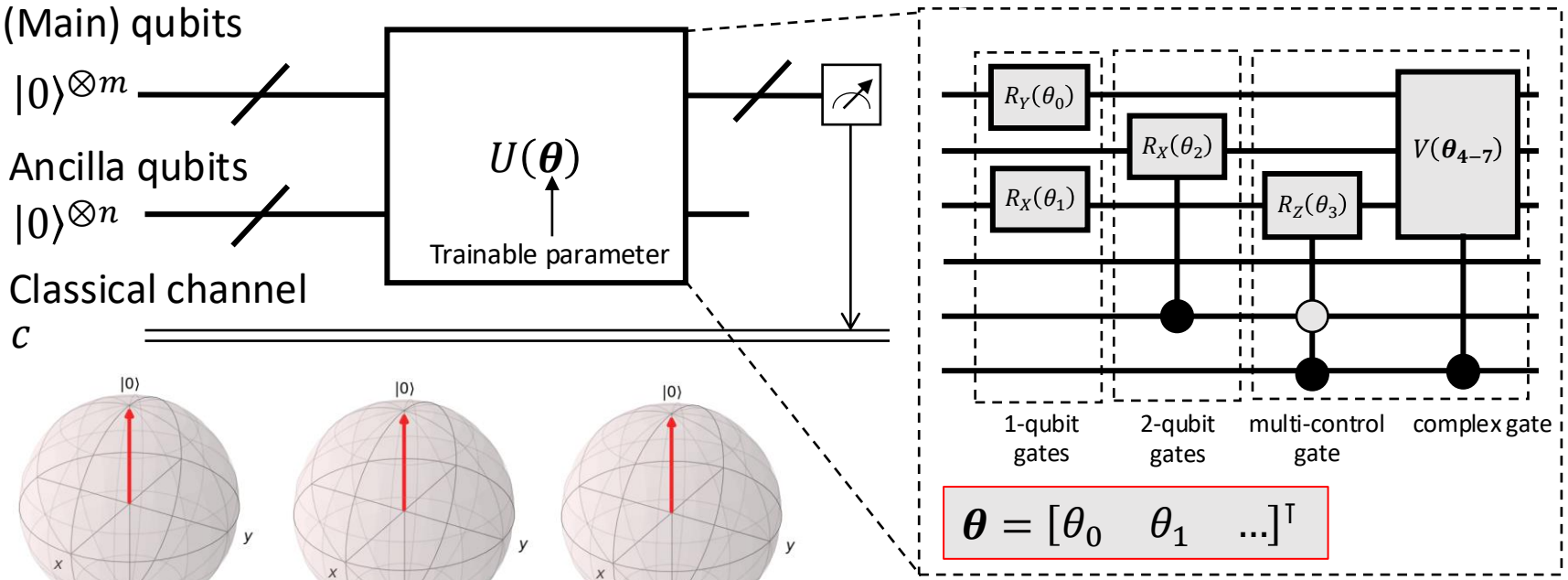
b) Quantum analogue version of a deep learning model

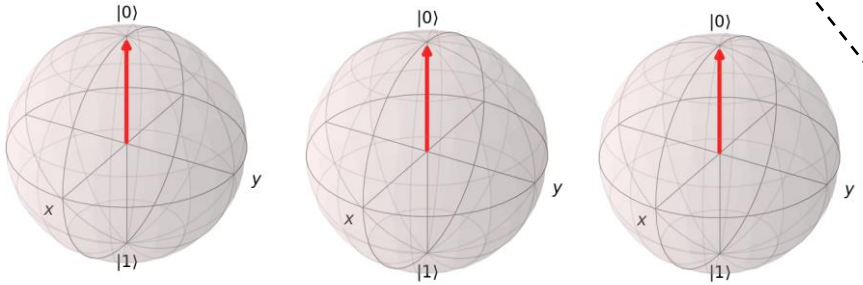
- QuNN, QCNN
- QGNN
- QGAN

c) Quantum learning

→ I don’t know this type

3.1. Parameterized quantum circuit (PQC)



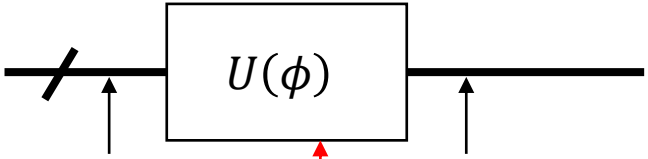


Gate $R_i(\theta)$ rotate state θ° about the i-axis where $i \in \{X, Y, Z\}$ and $\theta \in [0,2\pi]$

[3.1.1] Benedetti, M et al, Parameterized quantum circuits as machine learning models. Quantum Science and Technology, 4(4), 043001 (2019)

3.2. Encoding

Quantum state preparation



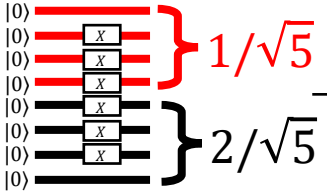
$|\psi_0\rangle = |0\rangle^{\otimes n}$ $|\psi'\rangle = U(\phi)|\psi_0\rangle$ with $|\psi'\rangle$ is the prepared/embedded state

How to prepare $U(\phi)$?

For example: $|\psi'\rangle = \begin{bmatrix} 1/\sqrt{5} & 2/\sqrt{5} \end{bmatrix}$

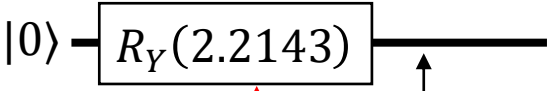
$0.447_{10} \approx 0.0111_2$

$0.8944_{10} \approx 0.1110_2$



$\longrightarrow$ $M2^n$ qubits

Basic encoding



$|\psi'\rangle = \begin{bmatrix} 1/\sqrt{5} \\ 2/\sqrt{5} \end{bmatrix}$ $\longrightarrow$ n qubits

How to find ϕ?

Amplitude encoding

[3.2.1] Schuld, M., & Petruccione, F. (2018). Supervised learning with quantum computers. *Quantum science and technology*, *17*.

3.2. Encoding

Diff encoders

Encoder / Feature	Threshold Encoding - **TE**	Flexible Representation - **FRQI**	Novel enhanced quantum - **NEQR** **[3.2.2]**	Amplitude Encoding - **AE** **[3.2.3]** & **Quantum RAM** **[3.2.4]**	Divide – and - Conquer Encoder - **DCE** **[3.2.5]**	Universal Compilation – based variational quantum- **UC – VQA** **[3.2.6]**
#qubit	$\mathcal{O}(2^{2n})$	$\mathcal{O}(2n+1)$	$\mathcal{O}(2n+1)$	$\mathcal{O}(n)$	$\mathcal{O}(2^{2n})$	$\boldsymbol{\mathcal{O}(n)}$
Circuit depth	$\mathcal{O}(2^{2n})$	$\mathcal{O}(2^{4n})$	$\mathcal{O}(2^{2n})$	$\mathcal{O}(2^{2n})$	$\mathcal{O}(k)$	$\boldsymbol{\mathcal{O}(\mathrm{poly}(n))} + \boldsymbol{\mathcal{O}(\mathrm{poly}(n))}$

List of encoders

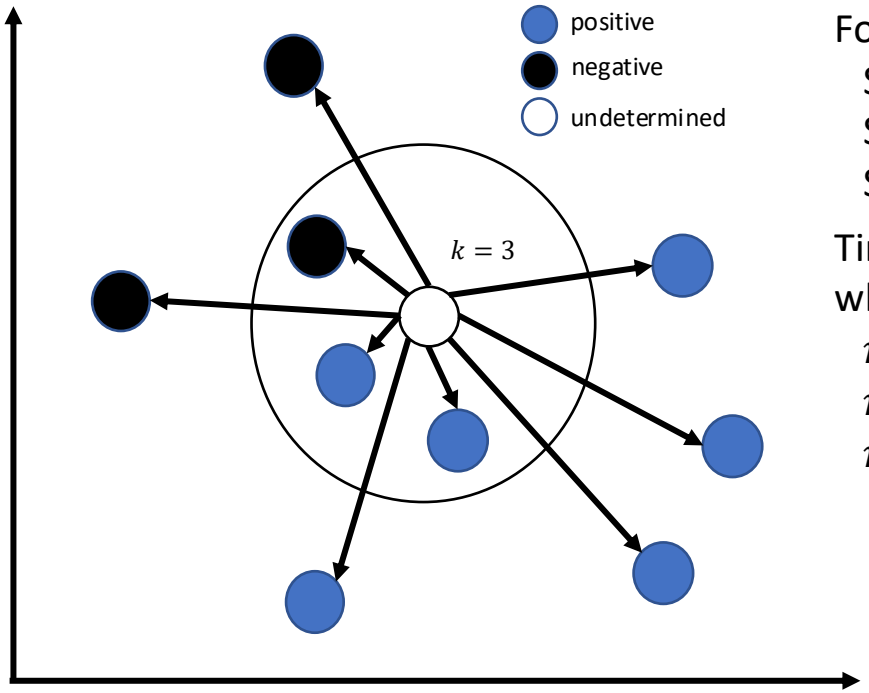
Circuit depth

Resource for $\boldsymbol{\theta_\phi}$

[3.2.2] Zhang, Y., Lu, K., Gao, Y., & Wang, M, NEQR: a novel enhanced quantum representation of digital images. Quantum information processing, 12, 2833-2860 (2013).
[3.2.3] M. Weigold et al, Data encoding patterns for quantum computing, in Proceedings of the 27th Conference on Pattern Languages of Programs PLoP '20, (The Hillside Group, USA, 2022).
[3.2.4] V. Giovannetti et al, Quantum random access memory, Phys. Rev. Lett. 100 (Apr 2008) p. 160501.
[3.2.5] I. F. Araujo et al, A divide-and-conquer algorithm for quantum state preparation, Scientific Reports 11 (Mar 2021) p. 6329.
[3.2.6] Hai, V. T., & Ho, L. B. (2023). Universal compilation for quantum state tomography. Scientific Reports, 13(1), 3750

3.3. QML algorithms

QkNN



For each test data point a:

Step 1: Calculate $\mathrm{Ds_a} = dist(a, x_i)\ \forall\ i = 1..n$

Step 2: Sort and get k smallest dist in $\mathrm{Ds_a}$.

Step 3: Get k corresponding label and do major vote

Time complexity: $\mathcal{O}(mnn_{\mathrm{test}})$

where:

m: dimensional of x_i

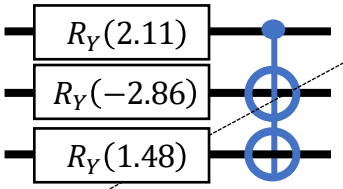
n: # train data points

n_{test}: # test data points

3.3. QML algorithms

QkNN

$$x = \begin{bmatrix}1\\0\\2\\3\end{bmatrix} \xrightarrow{\text{Normalized}} \hat{x} = \begin{bmatrix}1/\sqrt{14}\\0\\2/\sqrt{14}\\3/\sqrt{14}\end{bmatrix} \xrightarrow{\text{Encoded}}$$



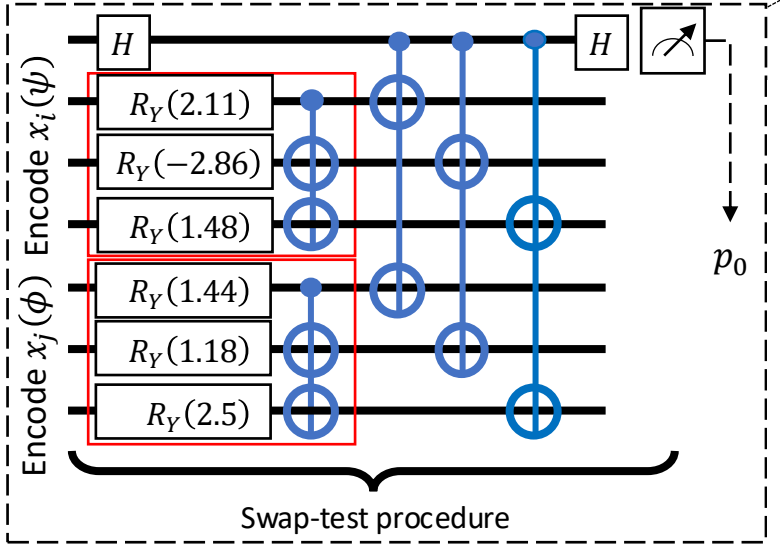
Fredkin-gate ≡ SWAP

For each test data point a:

Step 1: Calculate
$Ds_a = dist(a, x_i)\ \forall\ i = 1..n$

Step 2: Sort and get k smallest dist in Ds_a.

Step 3: Get k corresponding label and major vote



$$|0\rangle|\psi\rangle|\phi\rangle \xrightarrow{H_0} \frac{1}{\sqrt{2}}(|0\rangle|\psi\rangle|\phi\rangle + |1\rangle|\psi\rangle|\phi\rangle)$$

$$\xrightarrow{\{SWAP_{i,j}\}} \frac{1}{\sqrt{2}}(|0\rangle|\psi\rangle|\phi\rangle + |1\rangle|\phi\rangle|\psi\rangle)$$

$$\xrightarrow{H_0} \frac{1}{2}(|0\rangle|\psi\rangle|\phi\rangle + |1\rangle|\phi\rangle|\psi\rangle) + \frac{1}{2}(|0\rangle|\phi\rangle|\psi\rangle - |1\rangle|\psi\rangle|\phi\rangle)$$

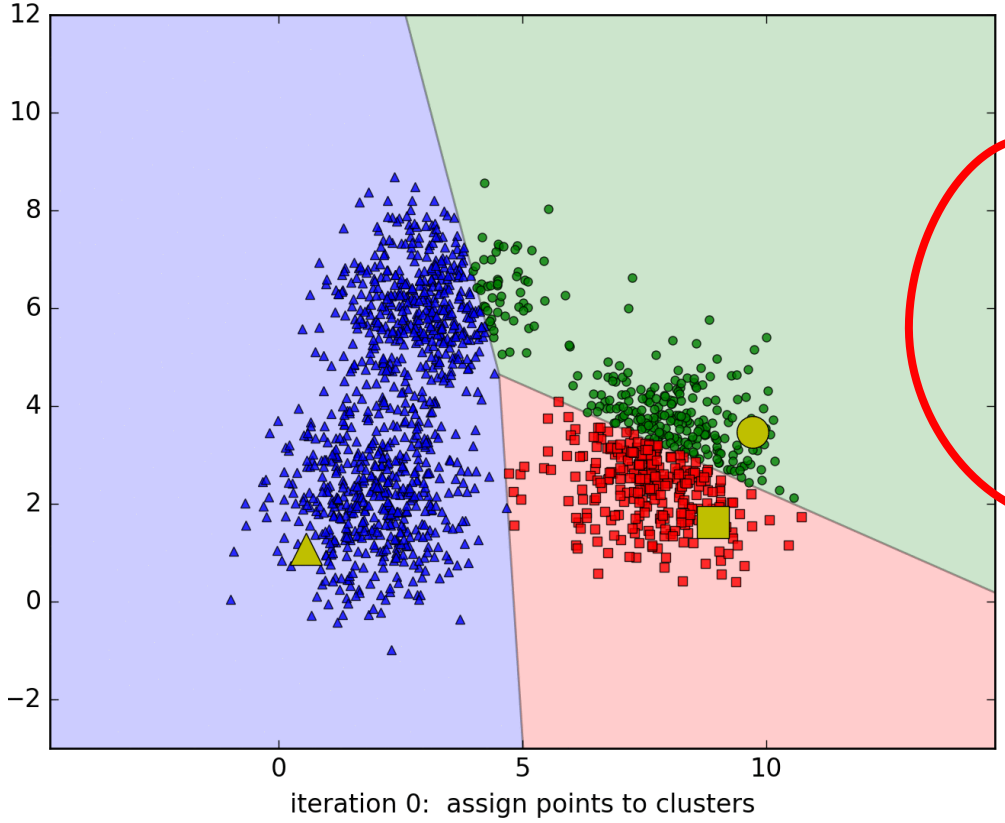
So $p_0 = \frac{1}{2}(\langle\phi|\langle\psi| + \langle\psi|\langle\phi|)\frac{1}{2}(|\phi\rangle|\psi\rangle + |\psi\rangle|\phi\rangle)$

$$\Rightarrow \boxed{|\langle\psi|\phi\rangle| = 2p_0 - 1}$$

[3.3.1] Vu Tuan Hai, Phan Hoang Chuong and Pham The Bao (2022). New approach of KNN Algorithm in quantum computing based on new design of quantum circuits. *Informatica*, *46*(5).

3.3. QML algorithms

k-means

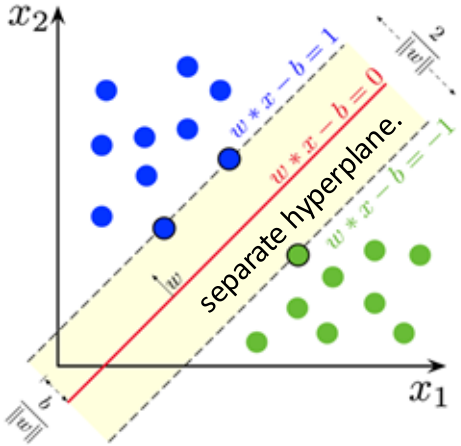


Step 1. Select k random centers.
Step 2. Assign each data point to the closest center.
If the assigned labels do not change, then stop.
Step 3. New center = average of all assigned points
Step 4. Return to step 2.

Swap-test procedure can be applied again!

3.3. QML algorithms

Quantum SVM



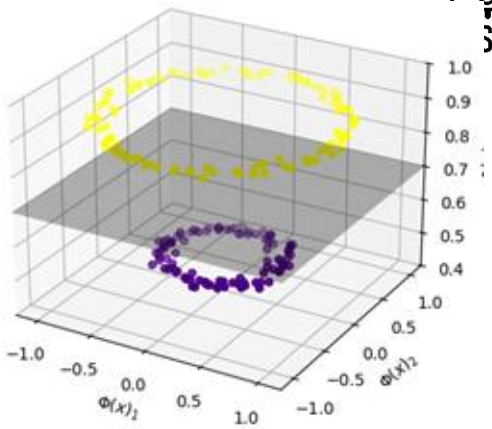
$\Rightarrow$

Inseparable data

$x = [x_1 \quad x_2]^\intercal$

Kernel method

$\Rightarrow$



Separable data

$\phi(x) = [x_1 \quad x_2 \quad x_1 x_2]^\intercal$

We have:

parameters

$$f^*(x) = \mathbf{sgn}(\boxed{w} \cdot x \pm \boxed{b}),$$

with target (w^*, b^*):

$$\boxed{\min_{w,b,c} \frac{1}{2}\|w\|_2^2 + C\sum_{i=1}^{n} \epsilon_i}$$

$$\boxed{\max_{\beta} L(\beta) = \sum_{i=1}^{n} \beta_i - \frac{1}{2}\sum_{i=1}^{n}\sum_{j=1}^{n} \beta_i \beta_j y_i y_j \boxed{\phi(x_i)^\intercal \phi(x_j)}}$$

Inner product

$\mathcal{O}(M * ?)$ comple

$$K(x_i, x_j) \equiv \phi(x_i)^\intercal \phi(x_j)$$

Kernel function is faster!

3.3. QML algorithms

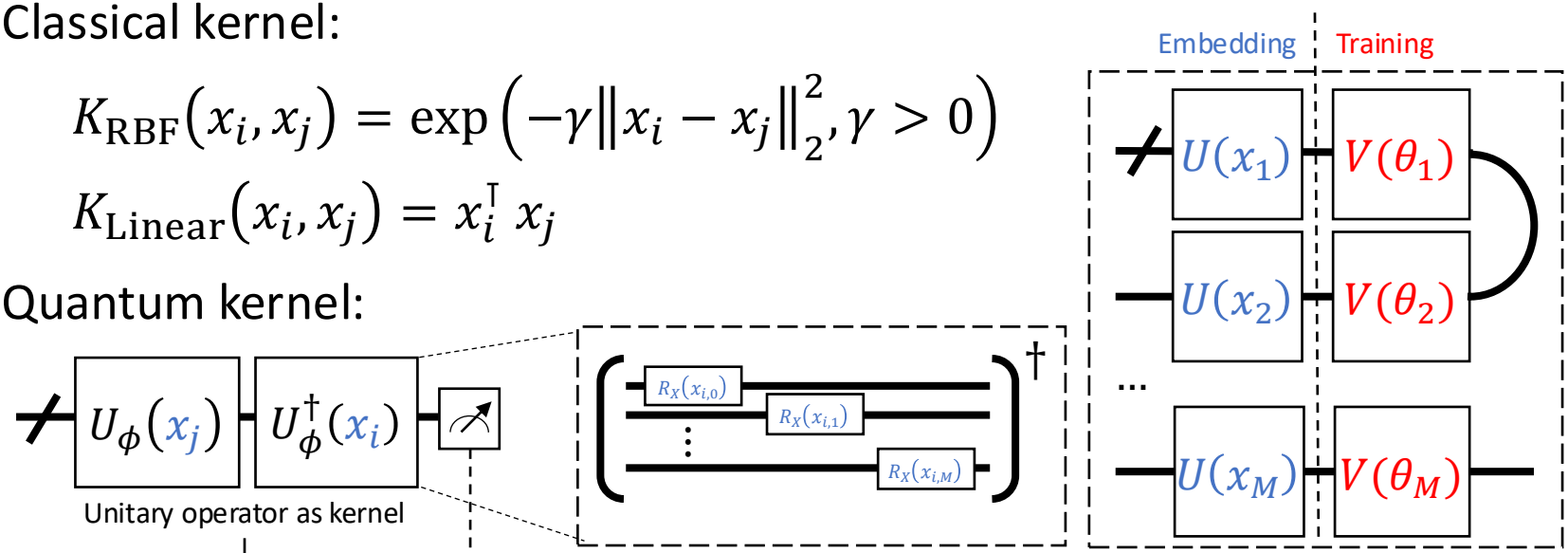
Quantum SVM

Classical kernel:

$$K_{\mathrm{RBF}}(x_i, x_j) = \exp\left(-\gamma \|x_i - x_j\|_2^2, \gamma > 0\right)$$

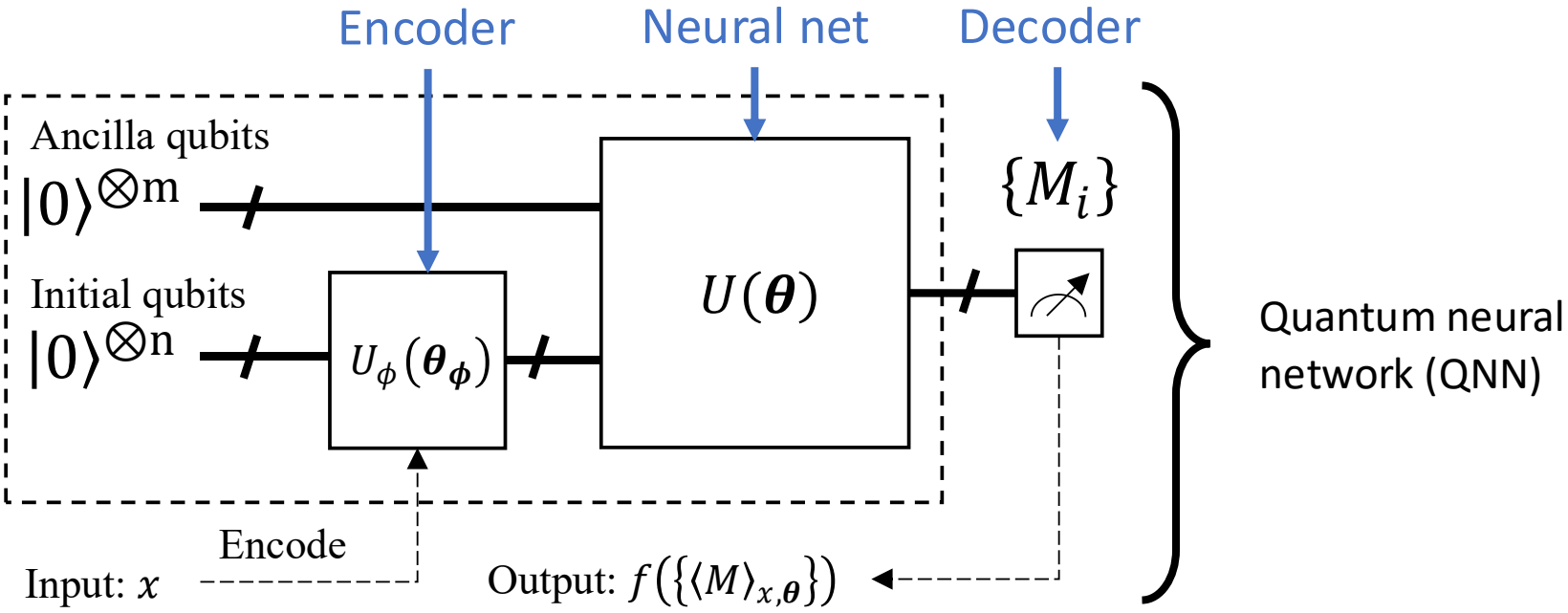
$$K_{\mathrm{Linear}}(x_i, x_j) = x_i^{\mathsf{T}}\, x_j$$

Quantum kernel:



$$p_0 = \underbrace{\left|\langle U_\phi(x_j) | U_\phi(x_i)\rangle\right|^2}_{\text{“Nature” inner product}}$$

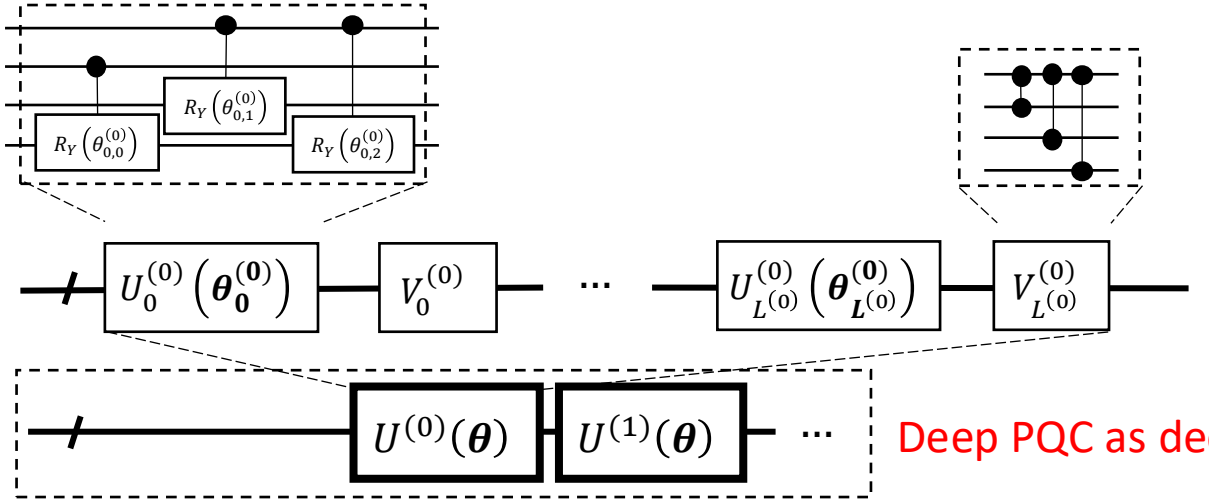
3.4. Quantum deep learning



[3.4.1] Benedetti, M et al, Parameterized quantum circuits as machine learning models. Quantum Science and Technology, 4(4), 043001 (2019)

3.4. Quantum deep learning

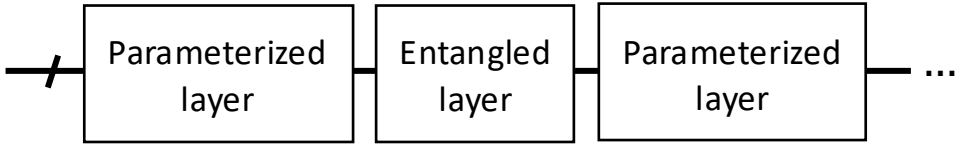
QNN



Deep PQC as deep QNN

Note that $U(\theta, \lambda, \phi) \equiv \left(R_X(\theta) \; R_Y(\theta) \; R_Z(\theta) \right) \times K$

Then



Successive parameterized gates are meaningless

Entangled gates (CX. MCX, Control-U, ...) should be appended in the middle

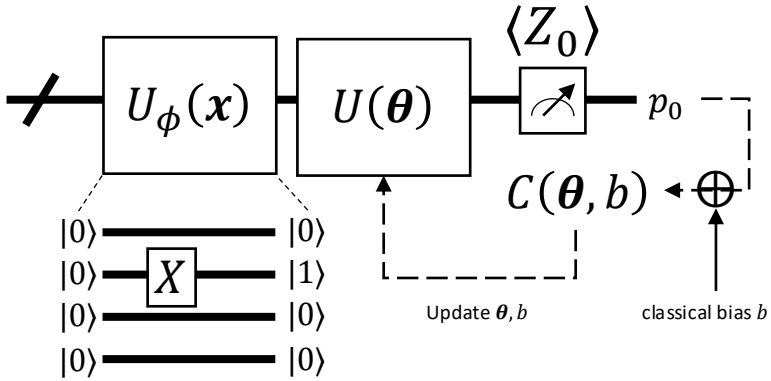
3.4. Quantum deep learning

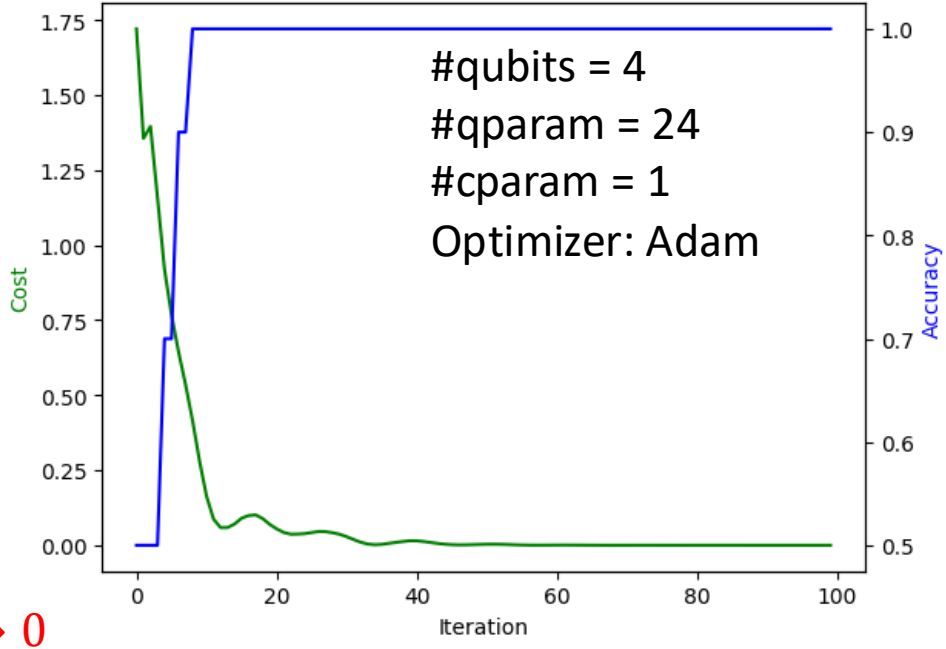
QNN

Attribute	**Classical deep learning**	**Quantum deep learning**
Main concept	Neural network	Parameterized quantum circuit (PQC)
Parameter	Weight $\{w_i\}$ in nodes	Angle value $\{\theta_\mathrm{i}\}$ in rotation gates $\{R_X, R_Y, R_Z\}$
Gradient	Backpropagation	General parameter-shift rule and gradient tensor metric
Optimizer	Classical optimizer	Classical optimizer + gradient-free optimizer
Loss	Various type	Associated with expectation values

3.4. Quantum deep learning

Variational classifier for simple classification problem





Problem:

Simulate $f: x \in \{0,1\}^{\otimes n} \rightarrow y = \bigoplus_i x_i$

Example: $f(|0100\rangle) = 1$

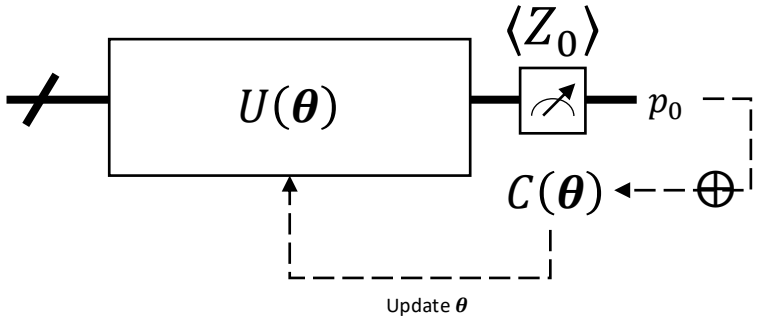
Cost function $C(\boldsymbol{\theta}, b) = \frac{1}{N}\sum(\hat{y}_i - y_i)^2 \rightarrow 0$

Accuracy metric $A = \frac{1}{N}\sum \mathbb{I}(|y_i - \hat{y}_i| < \tau) \rightarrow 1$

[3.4.2] https://pennylane.ai/qml/demos/tutorial_variational_classifier

3.4. Quantum deep learning

Variational classifier for ordinary differential equation (ODE)



Solve ODE: $y'(x) = -0.5\sin(x) + 0.5$.

Then find $y(x)$ $\left(y(x) = \frac{1}{2}(\cos(x) + x)\right)$.

Assume that $U(\boldsymbol{\theta}, x) \equiv y(x)$, then:

$$y'(x) = \frac{dU}{dx} = \frac{\left(U(\boldsymbol{\theta}, x+\epsilon) - U(\boldsymbol{\theta}, x-\epsilon)\right)}{2\epsilon}$$

We **simulate** $y(x)$ via $U(\boldsymbol{\theta}, x)$

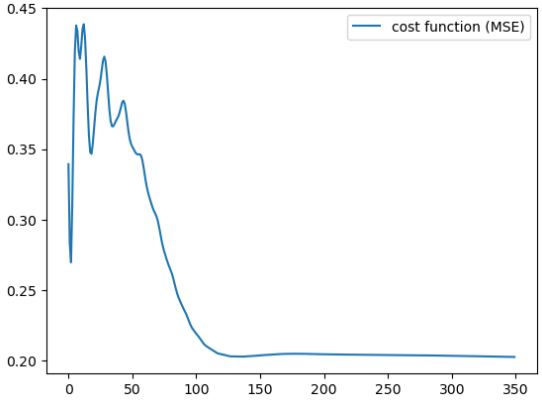
Cost function:

$$C(\boldsymbol{\theta}) = [U(\boldsymbol{\theta}, x_0) - y_0]^2 + \sum_j \left(\frac{dU(\boldsymbol{\theta}, x_j)}{dx} - y'(x_j)\right)^2$$

Sampling from y'

$$\{\partial C(\boldsymbol{\theta}) / \partial \theta_j\}$$

$$\min_{\boldsymbol{\theta}} C(\boldsymbol{\theta})$$



[*] Results from Nguyen Quoc Chuong with Python package Qsun

3.4. Quantum deep learning

Variational classifier for quantum battery optimization

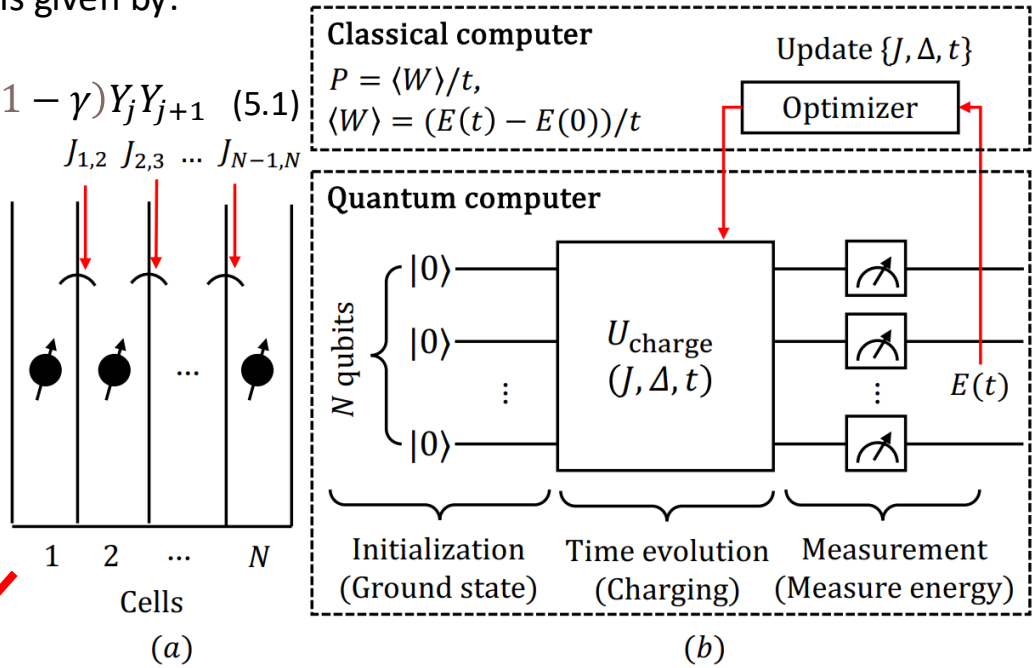
A quantum battery is given in the ground state of a Hamiltonian $H_0(h) = -h\sum_{j=1}^{N} Z_j$. The **charging Hamiltonian** is given by:

$$H_{\text{charge}}(J,\Delta,\gamma) = \sum_{j=1}^{N-1} J_j h(1+\gamma)X_jX_{j+1} + (1-\gamma)Y_jY_{j+1} \quad (5.1)$$

$$-\sum_{j=1}^{N-1} \Delta_j Z_j Z_{j+1} + H_0(h)$$

Optimizing $\{\boldsymbol{J},\boldsymbol{\Delta}\}$ mean we have a better battery

How to optimize $\{\boldsymbol{J},\boldsymbol{\Delta}\}$?



(a) A model of N-cell quantum battery.
(b) A quantum machine learning algorithm for optimizing battery's power.

3.4. Quantum deep learning

Variational classifier for quantum battery optimization

Classical | Quantum

t: charging time

$|\psi(0)\rangle$: initial ground state

After charging, we have power ($\boldsymbol{P}$) (performance of battery) :

$$\langle \boldsymbol{W} \rangle = \langle \boldsymbol{\psi(t)} | \boldsymbol{H_0} | \boldsymbol{\psi(t)} \rangle - \langle \boldsymbol{\psi(0)} | \boldsymbol{H_0} | \boldsymbol{\psi(0)} \rangle$$

$$\boldsymbol{\psi(t)} = \boldsymbol{e^{-iHt}} | \boldsymbol{\psi(0)} \rangle \qquad (5.2)$$

Calculate directly

$$\boldsymbol{P} = \langle \boldsymbol{W} \rangle / \boldsymbol{t}$$

Can be found after survey $t \in [0, T]$ with $t^{k+1} \leftarrow t^k + \Delta_t$

$$\boldsymbol{P_{max}} > \boldsymbol{P} \; \forall \boldsymbol{t}?$$

We want to find "max" of $P_{\max}$

$$\boldsymbol{P^*_{max}} > \boldsymbol{P_{max}} \; \forall \boldsymbol{J}, \boldsymbol{\Delta} \longrightarrow \boldsymbol{\mathcal{L}(J, \Delta, \gamma)} = -|\boldsymbol{P}_{\max}|$$

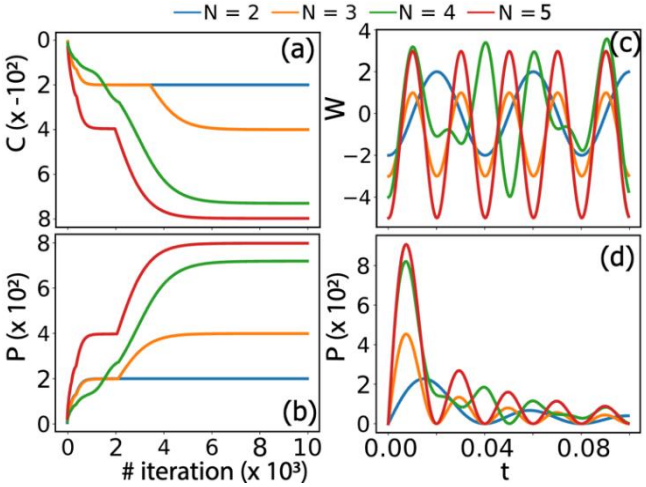
Calculate gradient

$$\boldsymbol{\nabla \mathcal{L}}$$

Update new $\{J, \Delta\}$ by classical optimizer

$$(\boldsymbol{J}|\boldsymbol{\Delta})^{t+1} = (\boldsymbol{J}|\boldsymbol{\Delta})^{t} - \boldsymbol{\alpha \nabla \mathcal{L}}$$

$$H_{\text{charge}} = \sum_{j=1}^{n-1} J_j \left(X_j X_{j+1} + Y_j Y_{j+1} \right)$$



(a) Plot of cost value

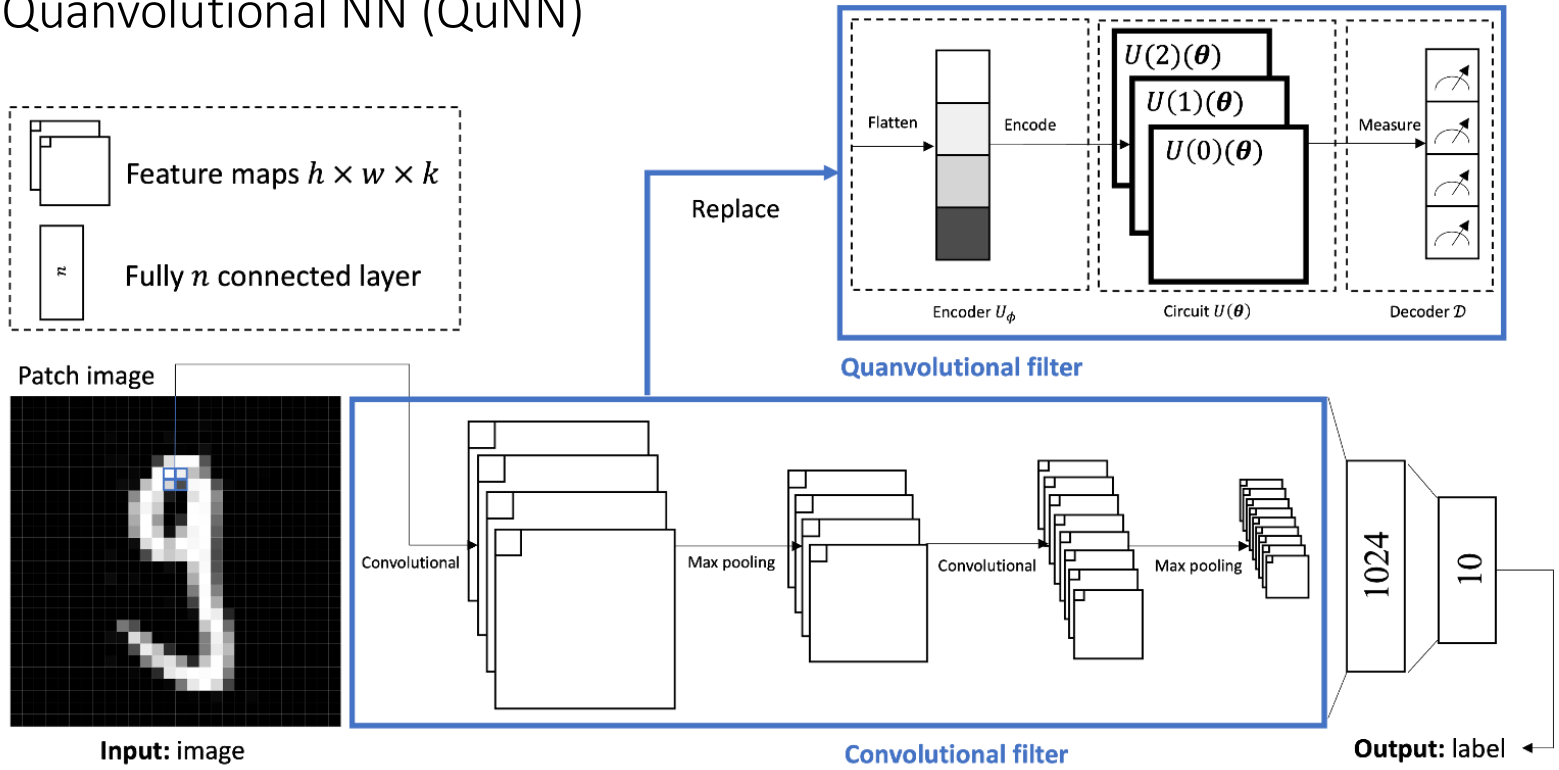
(b) Plot of power versus the iteration.

(c) Energy as a function of time.

(d) Power as a function of time.

3.4.1. Quantum deep learning

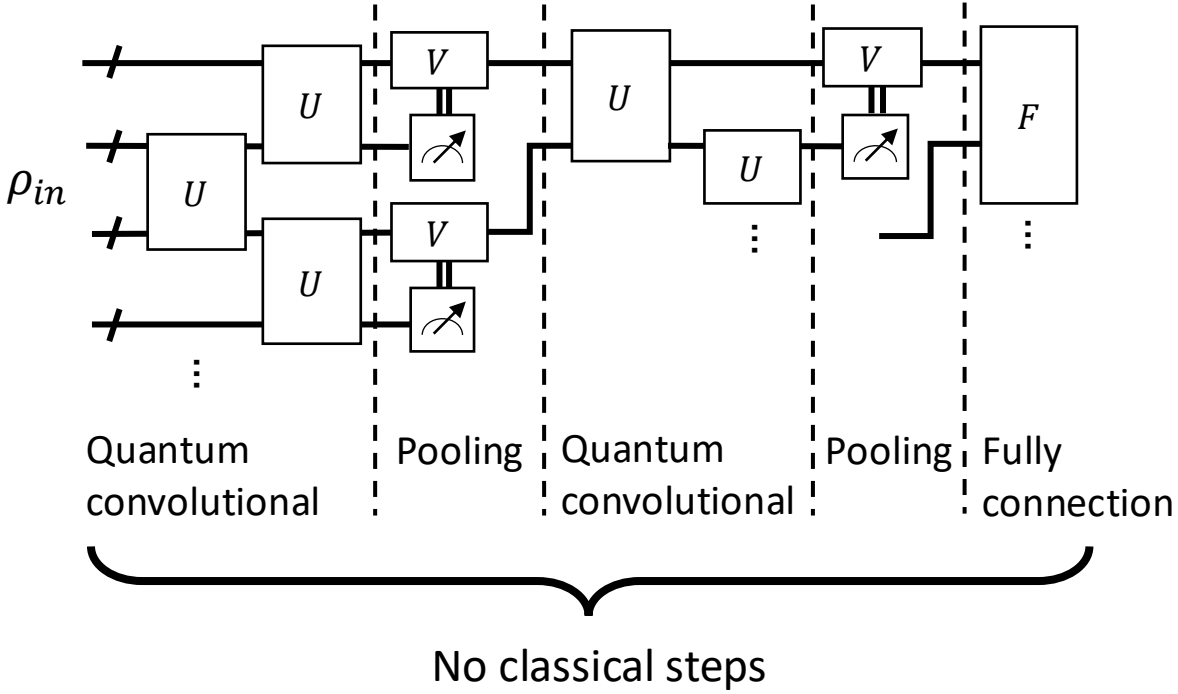
Quanvolutional NN (QuNN)



[3.4.2] Y. Lecun et al, Gradient-based learning applied to document recognition, Proceedings of the IEEE 86(11) (1998) 2278–2324
[3.4.3] T. Hur et al, Quantum convolutional neural network for classical data classification, Quantum Machine Intelligence4 (Feb 2022) p. 3.

3.4.2. Quantum deep learning

Quantum Convolution NN (QCNN)



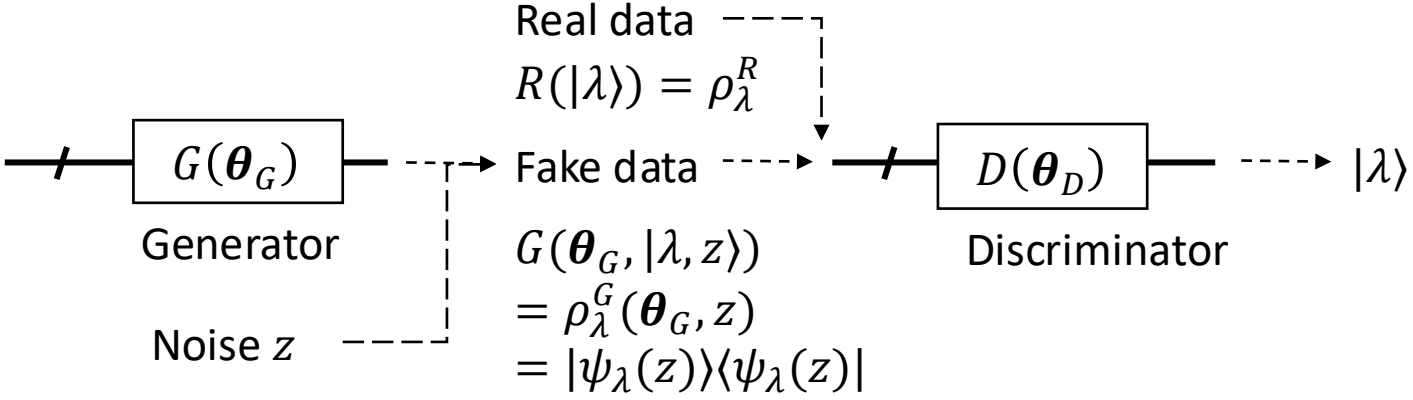
[3.4.5] Cong, I., Choi, S. & Lukin, M.D. Quantum convolutional neural networks. Nat. Phys. 15, 1273–1278 (2019). https://doi.org/10.1038/s41567-019-0648-8

3.4.3. Quantum deep learning

Quantum Generative adversarial net (QGAN)

Assume real data comes from the fixed distribution $p_R(x)$ generated by process R. Adversarial task:

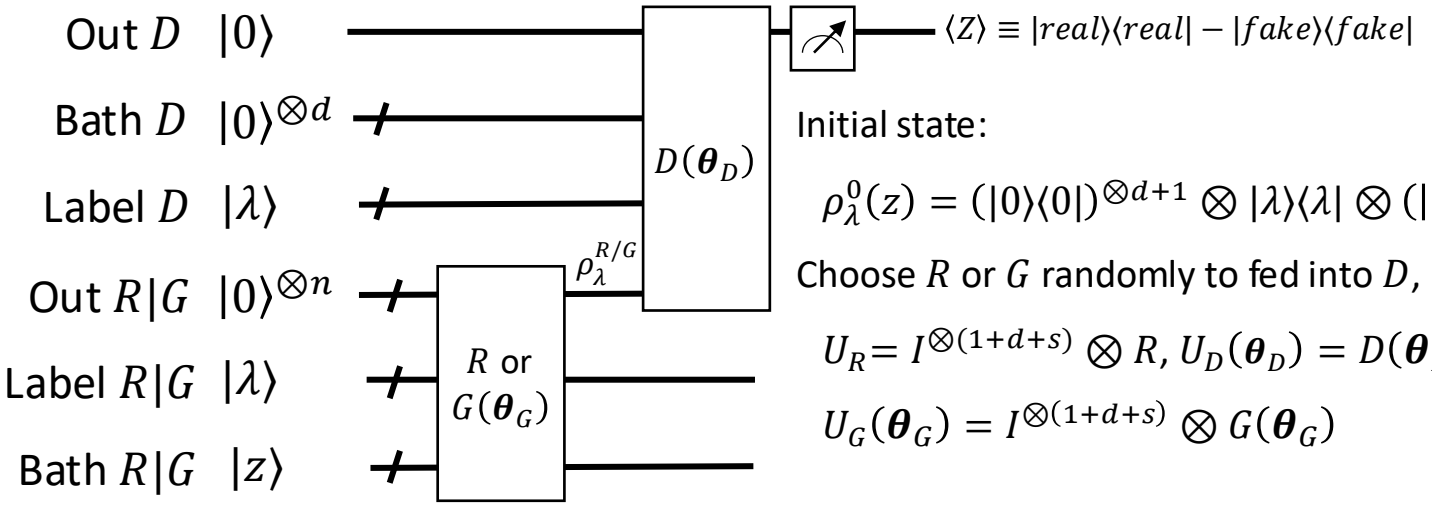
$$\min_{\boldsymbol{\theta}_G} \max_{\boldsymbol{\theta}_D} \sum_{\lambda=1} \Pr\Big(\big(D(\boldsymbol{\theta}_D, |\lambda\rangle, R(|\lambda\rangle)\big) = |\text{real}\rangle\big) \cap \big(D(\boldsymbol{\theta}_D, |\lambda\rangle, G(\boldsymbol{\theta}_G, |\lambda, z\rangle)\big) = |\text{fake}\rangle\big)\Big)$$



[3.4.6] https://pennylane.ai/qml/demos/tutorial_QGAN.html

3.4.3. Quantum deep learning

Quantum Generative adversarial net (QGAN)



Initial state:

$$\rho_\lambda^0(z) = (|0\rangle\langle 0|)^{\otimes d+1} \otimes |\lambda\rangle\langle\lambda| \otimes (|0\rangle\langle 0|)^{\otimes n} \otimes |\lambda\rangle\langle\lambda| \otimes |z\rangle\langle z|$$

Choose R or G randomly to fed into D, before fed:

$$U_R = I^{\otimes(1+d+s)} \otimes R,\ U_D(\boldsymbol{\theta}_D) = D(\boldsymbol{\theta}_D) \otimes I^m$$

$$U_G(\boldsymbol{\theta}_G) = I^{\otimes(1+d+s)} \otimes G(\boldsymbol{\theta}_G)$$

After fed:

$$\rho_\lambda^R = U_R \rho_\lambda^0(z) U_R^\dagger$$

$$\rho_\lambda^G(\boldsymbol{\theta}_G, z) = U_G(\boldsymbol{\theta}_G)\rho_\lambda^0(z)U_G^\dagger(\boldsymbol{\theta}_G)$$

After classify:

$$\rho_\lambda^{DR}(\boldsymbol{\theta}_D) = U_D(\boldsymbol{\theta}_D)\rho_\lambda^R U_D^\dagger(\boldsymbol{\theta}_D)$$

$$\rho_\lambda^{DG}(\boldsymbol{\theta}_D, \boldsymbol{\theta}_G) = U_D(\boldsymbol{\theta}_D)\rho_\lambda^G(\boldsymbol{\theta}_G, z)U_D^\dagger(\boldsymbol{\theta}_D)$$

[3.4.7] Dallaire-Demers, P. L., & Killoran, N. (2018). Quantum generative adversarial networks. *Physical Review A*, *98*(1), 012324.

3.4.3. Quantum deep learning

Quantum Generative adversarial net (QGAN)

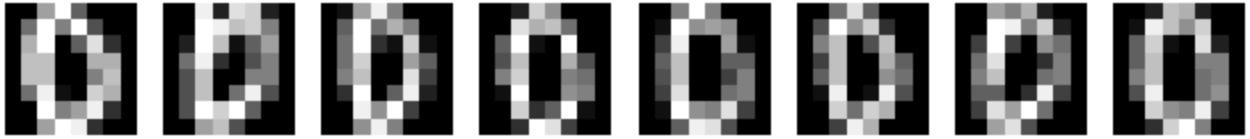
Quantum cost function

$$V(\boldsymbol{\theta}_D, \boldsymbol{\theta}_G) = \frac{1}{2} + \frac{1}{2\Lambda}\sum_{\lambda}\left(\cos^2(\phi)\mathrm{Tr}\left(Z\rho_{\lambda}^{DR}(\boldsymbol{\theta}_R)\right) - \sin^2(\phi)\mathrm{Tr}\left(Z\rho_{\lambda}^{DR}(\boldsymbol{\theta}_R, \boldsymbol{\theta}_D)\right)\right)$$

Here the angle ϕ parametrizes the probability that R or G is used as a source

Assume that $\phi = \frac{\pi}{4}$, mean p(R as source) $= p$(G as source)

$$V(\boldsymbol{\theta}_D, \boldsymbol{\theta}_G) = \frac{1}{2} + \frac{1}{4\Lambda}\sum_{\lambda}\mathrm{Tr}\left(\rho_{\lambda}^{DR}(\boldsymbol{\theta}_D) - \rho_{\lambda}^{DG}(\boldsymbol{\theta}_D, \boldsymbol{\theta}_G)\right)Z$$



3.4.3. Quantum deep learning

Quantum Generative adversarial net (QGAN)

In case G perfectly generates data source, $G(\boldsymbol{\theta}_G^*) = R$

$$\Rightarrow \Pr(Success\ D(\boldsymbol{\theta}_D)|\boldsymbol{\theta}_D) = \frac{1}{2} \Rightarrow \boxed{\nabla_{\boldsymbol{\theta}_D} V = 0, \nabla_{\boldsymbol{\theta}_G} V = 0}$$

Stop training

D is bounded by the purity function $C(\boldsymbol{\theta}_G) = \mathrm{tr}\big(\rho_R \rho_G(\boldsymbol{\theta}_G)\big)$:

$$\frac{1}{2} C(\boldsymbol{\theta}_G) \leq \Pr(Success\ D(\boldsymbol{\theta}_D)|\boldsymbol{\theta}_G) \leq 1 - \frac{1}{2} C(\boldsymbol{\theta}_G)$$

$\Rightarrow$ Increase gradient value when $\Pr \approx 1/2$

3.5. Additional information

Power of QML

We will focus on PQC which is the main computation in QML

Encoder: use a very small qubit ($\log_2 N$) to loading / encoding N – dimensional normalized x.

Operator: like classical NN, the structure of $U(\boldsymbol{\theta})$ decides the model extract feature / process information good or bad.

Decoder: quantum information → classical information.

3.5. Additional information

Challenge in QML field

- Standard architechture?
- Trainability, Expressibility
- Lack of experiment resources (not enough simulated #qubit to transform toy dataset → real dataset)
- Hype in QML: some applied QML models are actually worse than classical models.
- Other problems

[3.5.1] Joseph Bowles, Shahnawaz Ahmed, & Maria Schuld. (2024). Better than classical? The subtle art of benchmarking quantum machine learning models.
[3.5.2] Schuld, M., & Killoran, N. (2022). Is Quantum Advantage the Right Goal for Quantum Machine Learning?. PRX Quantum, 3, 030101.
[3.5.3] Jens Eisert, & John Preskill. (2025). Mind the gaps: The fraught road to quantum advantage.

Thank you for everything!